Confidence as Collateral

Strictly Proper Slashing for Accountable Automated Decisions

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This is a mechanism and a measured prototype, not a deployable protocol. Only the calibration head is zero-knowledge proved — its input logit is unverified, dispute resolution rests on an admin-appointed N-of-M committee whose collusion carries the authority a single key would, and an unvalidated collusion detector can withhold a claimant's payout. Section 8 states each limitation in full; Section 9 states what would close each one, and records that one closure made the problem harder rather than easier.

Abstract

Accountability mechanisms for automated decisions penalise error at a confidence-independent rate, so an operator is never punished for asserting 99% confidence in a decision it privately believes at 60%. We study slashing staked collateral by the **Brier score** over (reported confidence, adjudicated outcome). Because that score is strictly proper, expected loss is uniquely minimised when the operator reports its true subjective probability: honest confidence becomes the loss-minimising strategy rather than an assumption. We show the property survives fixed-point on-chain arithmetic, and prove that the protocol's slash cap must be 100% of stake — any lower cap destroys strict properness and makes maximal overconfidence optimal, so a prudential-looking parameter silently inverts the mechanism.

We report a measured prototype. Across 10 pinned seeds, temperature scaling cuts Expected Calibration Error from 0.1853 ± 0.0248 to 0.0870 ± 0.0218 on held-out UCI German Credit data, in every seed; an EZKL/halo2 circuit proves the calibration head in 2.09 s and verifies on-chain for 684,696 gas, flat across a 16,897× parameter increase. Five candidate enhancements were ablated against this baseline under an identical seed protocol: conformal prediction earns its place at no measurable proving cost, while a deeper base model is null and deep-ensemble epistemic uncertainty measurably *degrades* calibration.

Embedding the mechanism in an agentic decision market yields a participation condition and a sharp negative. Attaching a confidence attestation raises welfare only where verification costs less than the errors it screens out — a condition our measured L1 gas prices fail — and against a *single* buyer an optimally tuned flat-fraction bond achieves exactly the same welfare, so confidence elicitation buys nothing a correctly set parameter does not. The separation is real but specific: a bond implements one pooled threshold, while an attested confidence is a sufficient statistic each buyer applies its own threshold to, and across heterogeneous buyers an optimally tuned bond still forfeits 70.5% of the available gain.

The central result is negative and is the paper's main claim. A proved calibration head is **not** sufficient for a trustworthy attestation: the proof binds the map from logit to confidence and nothing else, so an operator that fabricates the logit obtains a proof that verifies. Two further measurements ran against the design. Deriving the unbonding period from statute rather than convenience gives 105 days, at which capital lockup — not the slash — becomes an operator's dominant cost. And the subgroup-calibration gap we set out to close proved *unmeasurable* at the first dataset's scale: the apparent effect is reproduced in full by a random partition, because binned ECE's small-sample bias at n = 68 exceeds the quantity being estimated. At 30,000 rows the same analysis resolves it — the effect is real but small, roughly 12% of aggregate ECE — which vindicates the null rather than overturning it, and still does not license an enforcement gate built on an estimator whose bias is twenty times the effect. We report both and did not build the on-chain component.

A last measurement concerns a component already shipped. The collusion detector reports an AUC of 0.998 and is wired to withhold a claimant's payout; at the one threshold the contract actually enforces it flags 0.94% of honest claimants on ring-free traffic, and one flagged claimant in five is honest at the easiest setting. Threshold-free metrics can make a system look validated for something it is not, and we recommend leaving the detector unattached. Every claimed weakness is executed as a passing test, so a failure means the threat model has drifted from the code.

Keywords: proper scoring rules, mechanism design, zero-knowledge machine learning, calibration, staking and slashing, agentic payments.

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1. Introduction

1.1 The regulatory setting

Automated credit decisions are moving into a compliance regime that demands records rather than assurances.

Under the EU AI Act, credit scoring is a high-risk application, and Article 26 places operational duties on **deployers** of such systems. Article 26(5) requires deployers to "monitor the operation of the high-risk AI system on the basis of the instructions for use" and to notify providers of risks or serious incidents without undue delay. Article 26(6) requires deployers to "keep the logs automatically generated by that high-risk AI system ... for a period appropriate to the intended purpose ... of at least six months." Article 86 establishes a right to explanation of individual decision-making. Article 26 becomes applicable on 2 December 2027 for most high-risk systems, with an extended deadline of 2 August 2028 for systems in employment, education, and access to essential services [1].

These provisions mandate *logging and explanation*. They do not establish that the logs are truthful, nor do they price the difference between a wrong decision made tentatively and a wrong decision made with asserted certainty. A deployer that logs a confidently wrong decision is compliant with Article 26(6).

A note on the Indian setting, correcting a common assumption. The RBI Digital Lending Directions, 2025 are frequently cited as imposing algorithmic transparency duties. They do not. The RBI Working Group recommended that regulated entities "document the rationale for algorithmic features and conduct regular algorithmic audits," but these recommendations were incorporated into neither the 2022 Guidelines nor the 2025 Directions, which remain silent on algorithmic fairness and transparency [2]. We therefore cite India as an example of a **regulatory gap** rather than a regulatory driver. We were unable to locate a provision in the 2025 Directions imposing model-explainability obligations, and we do not assert one exists.

Regulation currently creates demand for *auditable records of automated decisions*, and does not yet create demand for *calibrated confidence*. This paper argues that the latter is the more useful object; it does not claim the latter is presently required.

1.2 The gap in existing mechanisms

Decentralised protocols already slash staked collateral for misbehaviour, and already run decentralised claim adjudication. Neither weights the penalty by the reporter's own stated confidence.

- **Oracle slashing.** Chainlink slashes node operators for failing on-chain service-level obligations at a **fixed 700 LINK per valid alerting event** on the ETH/USD data feed, with the alerter rewarded 7,000 LINK [3]. The penalty is a constant: it is identical whether the operator was marginally or egregiously wrong, and oracles do not report confidence at all.
- **Decentralised insurance.** Nexus Mutual resolves claims by stake-weighted member voting, with claim assessors staking NXM and rewarded for voting with the outcome [4]. The mechanism adjudicates *whether a claim is valid*; it does not score how confident anyone was.
- **On-chain proper scoring rules.** These do exist. Foresight Arena (arXiv:2605.00420) scores AI forecasting agents on Polymarket questions using the Brier score, with commitments and scores recorded on Polygon PoS via smart contracts [5].

The precise delta is therefore narrower than "nothing like this exists," and we state it as such: **proper scoring rules have been applied on-chain to *reward* forecasters in prediction-market settings; we are not aware of prior work applying one to *slash staked collateral* of a model operator under dispute, with the scored quantity produced by a zero-knowledge-proved calibration step.** Foresight Arena is reward-based, involves no staked collateral at risk, no slashing, and no zero-knowledge proofs [5]. We claim novelty in the combination, not in any component.

1.3 Research questions and findings

The paper is organised around five questions. Each is stated with the evidence that settles it and the answer that evidence produced; two of the five answers are negative, and those are the paper's more useful results.

RQ1 — Incentive. *Does penalising a disputed decision by its Brier score make truthful confidence reporting the operator's loss-minimising strategy, and does that property survive fixed-point on-chain arithmetic?*

Answered affirmatively. Proposition 1 gives a unique expected-loss minimum at the operator's true subjective probability, and the deployed Solidity reproduces that minimum numerically at two distinct true probabilities with monotonicity fuzzed in both directions (§3.2, §7.8). Proposition 2 adds a condition the derivation does not survive without: the slash cap must be 100% of stake, because any lower cap makes a *boundary* report optimal and rewards maximal overconfidence (§3.3).

RQ2 — Feasibility. *Can the confidence-producing step be proved in zero knowledge and verified on-chain at a cost a per-decision workflow could absorb?*

Answered affirmatively, with the cost stated. 2.09 s to prove on a laptop CPU, 684,696 gas to verify on-chain — roughly 33× a plain transfer. The mechanism's own arithmetic is 543 gas of that, about 0.08%; essentially all cost is halo2 verification rather than confidence-weighted slashing (§7.3, §7.4).

RQ3 — Cost structure. *How does proving cost scale with the size of the proved head, and does that scaling favour proving a small calibration head over a full classifier?*

Answered, within a stated boundary. Proving cost is flat across a 16,897× parameter increase, so circuit *capacity* rather than parameter count is the binding constraint (§7.3). The boundary itself — where cost steps past logrows 15 — is unmeasured, and we say so rather than extrapolating.

RQ4 — Sufficiency. *Is a proved calibration head enough to make the resulting attestation trustworthy?*

Answered negatively, and this is the paper's central claim. The proof binds the map from logit to confidence and nothing else: an operator that fabricates the input logit obtains a proof that verifies and a chain that accepts it. Each component of the trust boundary is executed as a passing test rather than conceded in prose (§5.3, §8).

RQ5 — Sophistication. *Does a more capable model, or a stronger uncertainty quantifier, improve the quantity this mechanism actually prices?*

Answered largely negatively. Of five enhancements ablated against the shipped baseline on an identical seed protocol, conformal prediction earns its place, a deeper base model is null, and deep-ensemble epistemic uncertainty is actively harmful (§7.5, Appendix A). The one-parameter head survives every attempt made here to beat it.

RQ6 — Market value. *When is attaching a confidence attestation to a priced decision worth its verification cost, and does it dominate a simpler accountability bond?*

Answered conditionally, and partly against the mechanism. Attachment pays only where verification costs less than the screening value it buys — a condition our measured L1 gas prices fail (Proposition 4). Against a single buyer an optimally tuned flat bond matches it exactly (Proposition 5); the separation requires heterogeneous buyers, where a bond forfeits 70.5% of the gain (Proposition 6, §4).

1.4 Contributions

Mechanism.

1. A slashing rule built on a strictly proper scoring rule, with the properness argument derived in closed form (Proposition 1) and its scope conditions stated as part of the claim rather than as trailing caveats (§3.5).
2. **Proposition 2: the slash cap must be 100% of stake.** Capping the slash at $\varkappa$ preserves truthful reporting only if $\varkappa \geq \max(p, 1-p)$; below that the minimiser jumps to a boundary report, so a capped protocol *pays for* maximal overconfidence. This is limited liability's risk-shifting effect arriving through a parameter that looks prudential, and it also settles why the Brier score rather than the log score: the log score is unbounded, cannot be paid from finite stake, and truncating it triggers exactly this failure (§3.3).

Market.

1. A welfare model of a buyer–seller decision market (§4) yielding a **participation condition** — attaching an attestation pays iff its verification cost falls below the screening value it buys — and an **equivalence result**: against a single buyer an optimally tuned flat-fraction bond achieves identical welfare, so the mechanism's advantage is not over bonds in general but over *pooled* thresholds, and appears only under buyer heterogeneity (Propositions 3–6).

Systems.

1. A working implementation: fixed-point Brier arithmetic in Solidity at 543 gas, an EZKL/halo2 circuit over the calibration head, and an attestation contract that rejects any decision whose proof does not verify.
2. A measured evaluation over 10 pinned seeds, including a leakage control that comes out *worse than not calibrating at all*, and a head-capacity result reported at the strength the data supports — a majority effect at $p = 0.037$, not the uniform one a single run would have suggested.
3. A conformal-prediction layer carrying a distribution-free coverage guarantee, measured to fit the existing circuit budget, plus four further ablated enhancements of which two fail (§7.5, Appendix A).
4. Two on-chain surfaces — an operator reputation register and a content-addressed model-version registry — each placed explicitly in the trust tier it occupies rather than the one being on-chain might suggest, and a reference integration exporting the register to an external agent-identity standard (§5.3, §7.6).

Negative results, reported at the same weight as the positive ones.

1. An adversarial trust model (§5.3) in which every asserted weakness is demonstrated by an executing test, establishing that a proved calibration head is **not** sufficient for a trustworthy attestation (RQ4).

2. **The unbonding period, derived rather than assumed, is unaffordable.** A statutory derivation gives 105 days, at which capital lockup exceeds the expected slash by roughly 8× and becomes the dominant cost of participating — weakening this paper's own incentive argument (§8.3).
3. **A methodological caution on subgroup calibration.** The subgroup gap we set out to close is reproduced in full by a random partition of the same group sizes (Wilcoxon $p = 0.92$), because binned ECE is biased upward at small n: a model calibrated by construction scores 0.1188 at $n = 68$, above the aggregate ECE being compared against. Within-group ECE gates on datasets of this scale would fire on noise, so we report the null and did not build the on-chain component (§8.4).

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2. Related work

2.1 zkML and verifiable inference

Verifiable inference asks a narrow question: given a committed model and an input, can a party prove the output was computed correctly without revealing the model, the input, or both? The field divides by proof system and by what part of the pipeline is placed in circuit.

EZKL compiles computational graphs supplied in ONNX format into arithmetic circuits and produces zk-SNARK proofs of correct model execution, using a modified halo2 with Plonkish arithmetisation; anyone holding the verifying key can check the proof [6]. This prototype uses EZKL 23.0.5 directly. Related systems make different trade-offs along the same axis: zkCNN [11] specialises the proof system to convolutional structure and obtains large constant-factor gains at the cost of generality, while sumcheck-based approaches [12] target asymptotically cheaper proving for layered arithmetic circuits. The common finding across all of them is that cost tracks the proved subgraph, not the model's accuracy or its business importance.

That finding is the design constraint this work is built around, and it cuts in an unobvious direction. If proving cost scales with what is proved, then the right question is not "how do we prove the model?" but "what is the smallest computation whose correctness the mechanism actually needs?" For confidence-calibrated slashing the answer is the calibration head — one parameter — and §7.3 measures what that buys: cost flat across a 16,897× parameter increase, because fixed lookup and column overhead dominate at these sizes. The corollary, which §8.1 states as a limitation, is that everything outside the proved subgraph is unbound, and this paper is largely an argument about how much that omission costs.

2.2 Decentralised oracle staking and slashing

Staking mechanisms bond a reporter's capital against its behaviour, and differ in what triggers a loss and how large that loss is.

Chainlink's staking design uses on-chain service-level agreements, community alerting, and fixed-size slashing penalties [3]. The size is the salient detail: 700 LINK per valid alerting event on the ETH/USD feed, independent of how wrong the feed was. This is a deliberate simplification — a constant penalty needs no adjudication of degree — and it is exactly the property this mechanism replaces.

Nexus Mutual's claim assessment uses stake-weighted voting with a quorum, escalation to a full member vote, and reward for majority-aligned voting [4]. Kleros provides general decentralised arbitration: jurors are drawn from PNK stakers, self-select into sub-courts, vote on evidence, and are rewarded for voting with the majority under a Schelling-point incentive, with appeals heard by successively larger juries [7]. Both adjudicate a binary — was the claim valid, did the defendant breach — and neither scores the reporter's confidence, because in their settings the reporter does not state one.

The distinction worth drawing is between *slashing for misbehaviour* and *scoring a forecast*. Slashing asks whether a rule was broken; scoring asks how far a stated belief was from what happened. The literature is rich in the first and, on-chain, nearly silent on the second. Kleros remains the most plausible substitute for this prototype's committee (§9.2), and adopting it would inherit a body of work on juror incentives that this paper does not attempt to reproduce.

2.3 Calibration in machine learning

A model is calibrated when its stated confidences match observed frequencies: of the decisions it calls 80% likely, about 80% should hold. Accuracy and calibration are independent — a model can be accurate and overconfident, or inaccurate and perfectly calibrated — and this mechanism prices only the second.

Guo et al. show that modern neural networks are poorly calibrated, that this worsened as architectures grew, and that temperature scaling — a single-parameter extension of Platt scaling dividing logits by a learned $T > 0$ fitted on held-out data — is surprisingly effective at restoring it [8]. Because the map is monotone it cannot change any decision, so calibration is free in accuracy terms; §7.1 confirms accuracy is identical before and after, as it must be.

Measurement is itself contested. Expected Calibration Error depends on a binning scheme, and Nixon et al. show that equal-width binning is biased when predictions cluster, proposing adaptive schemes that equalise bin population [13]. This matters here more than usual, because ECE is not merely a reported diagnostic — it is the quantity the mechanism's value proposition rests on. The implementation is therefore explicit about its convention (§6.5) and is known-answer tested; the sensitivity of the headline result to the binning choice is not evaluated, and §8.6 records that as a limitation.

Temperature scaling is the primary method here for two reasons that happen to agree: it is empirically strong on small calibration sets, and a single-parameter affine map is the cheapest possible thing to arithmetise. §A.1 and §A.2 test whether more capable alternatives beat it, and find they do not.

2.4 Proper scoring rules

Brier introduced the mean-squared-error score for probability forecasts of binary outcomes, in a meteorological setting where forecasters had every incentive to hedge [9]. Gneiting and Raftery develop the general theory: a scoring rule is *proper* if the forecaster maximises expected score by issuing its true predictive distribution F over any $G \neq F$, and *strictly proper* if that maximum is unique [10]. They also give the Savage representation, under which every proper scoring rule corresponds to a convex function of the forecast, and the strictly proper rules to strictly convex ones — Brier corresponding to $F(p) = p^2$.

Strict properness is the sole property this mechanism requires, and Brier is not the only rule with it. The logarithmic score and the spherical score are also strictly proper, and the log score has a stronger claim in information theory, being the unique local proper rule [10]. Under a collateral constraint the ranking inverts, for a reason that is specific to slashing rather than to forecasting: the log score is unbounded, so it cannot be paid from a finite stake, and truncating it destroys the very properness it was chosen for. §3.3 states and proves that failure, and it is the reason the choice of Brier is a design decision rather than a default.

A separate strand applies proper scoring rules to elicit beliefs where no ground truth will ever arrive — peer prediction and Bayesian truth serum [14] — by scoring reports against other reports. That approach is directly relevant to the unobservable-counterfactual problem of §8.2, and is noted in §9.2 as an alternative to producing a ground truth that does not exist.

2.5 Agentic payments, and the trust layers being built around them

A body of work postdating this project's v0 targets the same problem from the payments side. It is summarised here and in full in `RELATED_WORK_V2.md`, which classifies each item as a substitute for this mechanism, a complement, or orthogonal to it.

The distribution of that classification is itself the finding, and it is not the one a crowded-field reflex would predict. **Almost all of this literature is about whether payment is correctly bound to service delivery, and almost none of it is about whether the delivered decision was any good.** The distinction is not rhetorical: it identifies which systems compete with this one and which merely sound as though they do.

Payment–execution binding. A402 [25] introduces atomic service channels that bind payment finalisation to service delivery via adaptor signatures, so payment settles only on release of execution-dependent secrets. Two independent security analyses [25, 26] show current x402 deployments fail at exactly this seam: optimistic execution before finality, settlement preemption, replay across the HTTP–chain boundary, and proxy-level header manipulation. All of it concerns whether the service *ran* and whether it was *paid for*. A402 closes correctly when a credit-scoring endpoint returns a score, and is indifferent to whether that score was 0.9 on an applicant who defaulted. These mechanisms and this one stack without overlapping.

One of those attacks does constrain the reference integration in `x402-middleware/`. Replay across the HTTP–chain boundary [26] applies to an attestation gate with the same force it applies to a payment: one valid attestation id can be presented against unlimited requests unless it is claimed, scoped, and expired. That mitigation is not implemented, and the middleware's documentation says so rather than leaving a reader to infer it.

Reputation and discovery. ERC-8004 [23] standardises on-chain agent identity and a reputation registry, deployed on Ethereum mainnet and 20+ networks. TraceRank [28] ranks services by propagating reputation through the payment graph. Neither observes decision quality: ERC-8004's feedback value is an arbitrary `int128` at caller-chosen precision, and TraceRank's signal is entirely endogenous to who paid whom. A service that is confidently wrong but popular with reputable payers ranks well.

The first empirical study of the deployed ERC-8004 ecosystem [24] measured what that permissiveness produces. Across Ethereum, BSC, and Base through May 2026: only 3%, 4%, and 15% of registrations expose a valid registration file with a live endpoint; 73.5%, 59.2%, and 90.6% of reviewers exhibit coordinated Sybil behaviour; and after removing Sybil-flagged feedback, 15.8%, 77.9%, and 86.8% of rated agents lose all valid feedback. The authors conclude the registry "cannot function as a trust signal": values are not commensurable, feedback is rarely grounded in verifiable interactions, and reputation is manipulable at minimal cost.

This is the strongest external evidence for the specific thing this mechanism does, and it must be cited as evidence rather than paraphrased as endorsement. Their three failures map onto three properties of the register in §5.1: the value is a fixed published function of (confidence, outcome); the write path is a resolved dispute against a staked operator; and manufacturing a favourable score requires not losing that dispute, so the cost of manipulation is the stake. `ERC8004ReputationAdapter.sol` writes Brier's EMA into that registry for exactly this reason — not to compete with the standard but to occupy a slot on it with a differently-provenanced number.

The caveat travels with the claim. Brier does not solve ungrounded reputation. It *relocates* the grounding assumption, from "anonymous reviewers are honest" to "the resolver committee is not colluding." That is a strictly smaller and more auditable assumption, and it is a real improvement — but it is not the elimination of trust, and §8.2 says what a colluding committee can still manufacture.

Reputation-priced access. ACHIVX's `@achivx/x402` middleware is a live product implementing reputation-gated pricing: it reads the agent's wallet from the payment header, looks up a trust level on a 0–5 scale, and applies a provider-configured price multiplier. Its seven dimensions — activity volume, payment reliability, supplier diversity, tenure, feedback score, dispute frequency, consistency — score the *buyer* on payment behaviour. None observes whether a decision the agent sold was right. The two compose: price by counterparty risk, gate on attested decision quality.

The one true substitute. HeLa Chain's accountability bonds [29] are this mechanism with the confidence term removed. Sponsors stake collateral for an agent, B(a) = Σ S_i, and "a fraction of B(a)" is slashed when a governance dispute resolves against the agent. The slashing condition mentions no confidence and no prediction quality. That flat fraction is precisely the confidence-independent penalty rate §1.2 identifies as the gap: an agent asserting 99% and one asserting 51% pay identically for the same wrong call, so nothing induces the 51%.

§4 models this comparison rather than asserting it, and the result is more qualified than this paragraph alone would suggest — against a single buyer an optimally tuned bond matches this mechanism exactly, and the separation appears only with heterogeneous buyers.

2.6 Positioning

The delta is narrow and the table states it precisely. *Conf* = the penalty is weighted by the reporter's own stated confidence; *Stake* = the reporter has collateral at risk; *ZK* = the scored quantity is produced by a zero-knowledge-proved computation; *Adjud.* = how the outcome is determined; *x402* = the system is deployed against the agentic-payments rail today.

Table 1: Positioning against the mechanisms this work draws on and the agentic- payments systems of §2.5. Feature attributions are from the cited sources and describe those systems as documented, not as re-implemented here.

Work	Conf	Stake	ZK	Adjud.	x402	Primary contribution
Chainlink staking [3]	–	yes	–	alerting	–	Fixed-size slashing on SLA breach
Nexus Mutual [4]	–	yes	–	member vote	–	Stake-weighted claim assessment
Kleros [7]	–	yes	–	juror pool	–	General decentralised arbitration
Foresight Arena [5]	yes	–	–	market	–	Brier-scored forecasting leaderboard
EZKL [6]	–	–	yes	–	–	ONNX-to-halo2 verifiable inference
ACHIVX	–	–	–	–	**yes**	Reputation-priced access, buyer-side
ERC-8004 [23]	–	–	–	client feedback	**yes**	Standard identity + reputation registries
TraceRank [28]	–	–	–	payment graph	**yes**	Sybil-resistant service discovery
HeLa bonds [29]	–	yes	–	DAO vote	–	Flat-fraction accountability bond
This work	**yes**	**yes**	**yes**	N-of-M	–	Confidence-weighted slashing, measured

Two things the table is meant to make hard to miss.

The final row is the only one carrying both *Conf* and *Stake*, and that conjunction is the novelty claim: of the mechanisms that stake collateral behind an agent, this is the only one whose slash is a strictly proper function of reported confidence; of the mechanisms that publish agent reputation, the only one whose write path requires an adjudicated loss.

It is also **not x402-native**, and the column exists so that shows rather than hides. ACHIVX, ERC-8004, and TraceRank are deployed against real agent traffic today; this is a prototype with an unaudited reference integration. On adoption the comparison is unfavourable, and no row is dominated on every axis.

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3. The mechanism

3.1 The slashing rule

Definition 1 (Confidence-calibrated slash). Let S be the collateral an operator has staked, $c \in [0,1]$ the confidence it reported that its decision was correct, and $o \in \{0,1\}$ the adjudicated outcome of a dispute over that decision. On an upheld dispute the contract slashes $slash = S \cdot (c - o)^2$, subject to a protocol maximum fraction of S.

The penalty is therefore the Brier score of a single forecast, denominated in stake. Everything that follows depends on one property of that score and on nothing else about it.

Table 2: Notation. Symbols are fixed here and used consistently throughout. Values in the final column are those of the deployed prototype.

Symbol	Meaning	Value
S	operator collateral at stake	demonstration values
c	reported confidence the decision was correct, in [0,1]	attested, WAD
o	adjudicated outcome of a dispute, 0 or 1	committee output
p	operator's private belief that the decision is correct	unobservable
L	base-model logit entering the calibration head	unverified input
T	temperature of the calibration head	3.47 ± 0.45
q	split-conformal quantile at level α	fitted off-chain
α	conformal miscoverage level	0.05, 0.10, 0.20
vk	halo2 verifying key identifying the proved head	one per version
τ	unbonding delay before a withdrawal may execute	7 days (floor 1 h)
N/M	resolver threshold over committee size	2 of 3

3.2 Why truthful reporting minimises expected loss

Assumption 1 (Truthful adjudication). The operator privately believes the decision is correct with probability p. A dispute, if opened, is adjudicated truthfully, so $o = 1$ with probability p; and the decision to open a dispute is independent of the reported c.

Proposition 1 (Strict properness of the Brier slash). *Under Assumption 1, an operator reporting c faces expected slash $E[slash] = S\ (\ (c - p)^2 + p(1 - p)\ )$. This is uniquely minimised at $c = p$, and every misreport costs $S(c - p)^2$ in expectation.*

Proof. The dispute is upheld with probability p, in which case $o = 1$ and the slash is $S(c - 1)^2$; otherwise $o = 0$ and the slash is Sc^2. Hence $E[slash] = S\ (\ p(c - 1)^2 + (1 - p)c^2\ ) = S\ (\ c^2 - 2pc + p\ )$. Completing the square in c gives $S\ (\ (c - p)^2 + p(1 - p)\ )$. The bracket is a strictly convex parabola in c whose only stationary point is $c = p$, and the residual $p(1 - p)$ does not depend on c, so the minimum is unique and the excess cost of reporting $c \neq p$ is exactly $S(c - p)^2$. □

Two consequences follow, and together they are the whole argument.

The unique minimum is at $c = p$. The expression is a strictly convex parabola in c, so any misreport in either direction costs the operator $S(c - p)^2$ in expectation. Overclaiming confidence is not merely unrewarded, it is priced quadratically, so shading the truth slightly is cheap while asserting near-certainty about a coin flip costs close to the entire stake. This is strict properness [10] instantiated as a collateral rule: the operator does not need to be honest, it needs only to be selfish.

Corollary 1 (Uncertainty is priced, and separately from dishonesty). *The residual $S \cdot p(1 - p)$ is irreducible: it is incurred whatever the operator reports, and is maximised at $p = 0.5$.*

An operator facing genuine uncertainty pays for it however honestly it reports. That is the economically correct reading: the term is the price of writing an opinion on a hard case. The mechanism therefore separates the cost of *being uncertain* from the cost of *misrepresenting uncertainty*, and only the second is avoidable by the operator — which is precisely the behaviour a slashing rule should reward.

The deployed contract computes this in fixed point rather than over the reals, so properness is re-established empirically against the shipped code rather than inherited from the algebra: §6.7 scans all 101 candidate reports at two distinct true probabilities and confirms the expected-loss minimiser is the true one in both, and monotonicity is fuzzed at 256 runs per direction.

3.3 Why the Brier score, and why the cap must be 100%

Brier is not the only strictly proper scoring rule, and the choice among them is usually treated as a matter of taste. Under a collateral constraint it is not: boundedness stops being an aesthetic property and becomes the difference between a mechanism that works and one that cannot be implemented.

The logarithmic score $-\log c$ (on $o = 1$) is strictly proper and is the standard alternative. It is also unbounded as $c \to 0$, so a slash defined from it is unbounded, and no finite stake can cover it. Any implementation must therefore truncate it — and truncation is exactly what the next result shows to be fatal. The spherical score is bounded but has no natural scale in units of stake. Brier is the proper rule whose range is already $[0,1]$, so "slash the whole stake" and "the worst possible score" coincide, and the cap never has to bind.

That last point is usually stated as a convenience. It is a correctness requirement.

Proposition 2 (Capping the slash destroys strict properness). *Let the contract slash $S \cdot \min((c - o)^2,\ \varkappa)$ for a cap $\varkappa \in (0,1]$. Truthful reporting minimises expected loss if and only if $\varkappa \geq \max(p,\ 1 - p)$. For $\varkappa < \max(p,\ 1 - p)$ the unique minimiser is a boundary report: $c = 0$ when $p < 1/2$, and $c = 1$ when $p > 1/2$.*

Proof. Write $E(c)$ for the expected slash in units of S. Truthful reporting yields $E(p) = p(1 - p)$, provided neither branch is capped at $c = p$. The two boundary reports yield $E(0) = p \cdot min(1, \varkappa) = p\varkappa$ and $E(1) = (1 - p)\ \varkappa$, since $\varkappa \leq 1$. Comparing, $E(0) < E(p)$ exactly when $p\varkappa < p(1 - p)$, that is when $\varkappa < 1 - p$; and $E(1) < E(p)$ exactly when $\varkappa < p$. So the truthful report is beaten by some boundary report precisely when $\varkappa < max(p,\ 1 - p)$, and survives otherwise. Since $max(p,\ 1 - p) \geq 1/2$, any cap below one half breaks properness at every $p \neq 1/2$. At $\varkappa = 1$ the constraint $\varkappa \geq max(p, 1 - p)$ holds for all p, the caps never bind, and Proposition 1 applies unchanged.

□

The direction of the failure is what makes this worth stating. A cap does not merely weaken the incentive; it inverts it. Capped liability protects the operator from the worst outcome of an extreme report, so extreme reports become cheap, and the mechanism starts paying for exactly the overconfidence it was built to price. This is the familiar risk-shifting effect of limited liability, arriving through a parameter that looks prudential.

The practical consequence is a deployment rule. `maxSlashBps` must be 10,000, and it is: `test_cap_atOneHundredPercent_preservesProperness` confirms the minimiser is the true probability at four values of p, while `test_cap_belowHalf_rewardsMaximalOverconfidence` shows a 50% cap sending an operator to $c = 0$ or $c = 1$, and `test_cap_propernessHoldsExactlyOnTheInterval` pins the boundary at $\varkappa = 0.75$. An operator lowering the cap to reduce exposure would silently convert the protocol into one that rewards the behaviour it exists to punish, so the constraint is enforced by a red test suite instead of a comment.

3.4 What the mechanism requires, and where else it applies

Definition 1 is stated over a (confidence, outcome) pair and a stake. Nothing in it mentions credit, and the properness argument of §3.2 uses no property of the decision beyond the three the definition names. It is therefore worth separating what the mechanism needs from what this particular deployment happens to supply.

The mechanism requires exactly three things:

1. **A scalar confidence** the operator commits to before the outcome is known.
2. **A binary outcome** that is eventually adjudicated.
3. **Collateral** the operator loses access to while a decision is disputable.

Any setting supplying those three admits the same rule, and the properness argument transfers unchanged. Candidate settings, in decreasing order of fit:

- **Price and data oracles.** Reporters already stake, and already have a natural notion of being wrong. What they do not currently do is report confidence at all (§1.2), which is a reporting-format change rather than a mechanism change. The outcome is observable, which removes the hardest problem this work has (§8.2).
- **Model marketplaces and inference markets.** A served prediction with an attached confidence, an escrowed bond, and a later ground-truth reveal is structurally identical to the credit case, with the advantage that the counterfactual is usually observed.
- **Automated underwriting and claims triage.** Same shape as credit, same unobservable-counterfactual problem for declines.
- **Agent action markets.** An agent that commits to a confidence before taking a consequential action fits the definition, but "the action was wrong" is frequently not binary, which breaks requirement 2.

The setting that does **not** fit, and which we flag because it is the one most often assumed to: free-form generation. It has no canonical scalar confidence, no binary ground truth, and no stable notion of the same decision across sampling seeds (§9.4).

Credit scoring was selected as the instantiation for three reasons: the regulatory pressure is concrete (§1.1), a standard public dataset exists, and the counterfactual problem is at its *hardest* there. A mechanism that survives the rejection case is not made worse by settings where outcomes are observed; the converse would not hold.

3.5 Scope conditions, stated as part of the claim

The derivation assumes the adjudicated outcome is truthful and that dispute selection is independent of the reported confidence. Neither condition is guaranteed by this prototype. Specifically:

1. **Input-logit provenance is unproven.** The circuit takes the base model's logit as an unverified public input. An operator that fabricates the logit obtains a proof that verifies. The proof binds the calibration step, not the pipeline (Figure 3, tier 1; `ThreatModel.t.sol::test_tier1_marginIsUnverifiedOperatorSuppliedInput`).
2. **Off-chain misreporting before proving is not prevented.** Nothing forces the operator to feed the circuit the logit its deployed model actually produced.
3. **Selective disputes break the properness argument.** If only rejections are ever disputed, the operator faces an asymmetric loss surface, and the dominant strategy under that surface is not analysed here.

The mechanism is proper; the *system* is proper only insofar as these conditions hold, and in v0 they do not. Closing conditions 1 and 3 is the substance of the open work in §9.

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4. An economic framework for agentic decision markets

§3 establishes that the slash elicits truthful confidence. It does not establish that anyone should want it. This section builds the market around Definition 1 and asks the question a buyer would: **what does attaching a Brier attestation to a priced decision buy, and when is it not worth the gas?**

The setting is the one x402 created. Autonomous agents now buy and sell individual decisions over HTTP — pay-per-inference, pay-per-feed, pay-per-risk-flag — settled in stablecoins with no prior relationship between the parties (§2.5). That is precisely a market in which the seller's confidence is private, unverifiable, and payoff-relevant to the buyer.

The results are stated in advance, including the one that does not favour the mechanism:

1. Under plain x402 the reported confidence carries **no** information, because reporting is costless (Proposition 3).
2. Attaching the Brier slash makes it informative, and the welfare gain is the **screening value** G — but only where the attestation costs less than G. At L1 gas prices with our measured parameters, it does not (Proposition 4).
3. Against a **single** buyer, an optimally tuned flat-fraction bond — HeLa's mechanism, §2.5 — matches Brier **exactly**. Brier's advantage there is that it requires no tuning, not that it achieves something a bond cannot.
4. The genuine separation appears only with **heterogeneous** buyers, where a bond implements one pooled threshold and a report is a sufficient statistic each buyer applies its own threshold

to (Proposition 5).

Every number below is produced by `scripts/22_market_model.py` and written to `artifacts/calibration/market_model.json`.

4.1 The interaction

A seller-agent holds a decision — approve or reject this applicant — and a private belief p that the decision is correct. A buyer-agent purchases it through an x402-gated endpoint at price ϱ and must then choose whether to act.

Definition 2 (Decision purchase game). A buyer with payoffs (V, K) acquires a decision and either acts or abstains. Acting on a correct decision yields V; acting on an incorrect one costs K; abstaining yields 0. The buyer observes only the seller's reported confidence c. The seller has private belief p and reports c at zero direct cost.

A buyer whose posterior that the decision is correct is π acts iff $\pi V - (1 - \pi)K > 0$, that is iff

$$pi \geq t = (K)/(V + K).$$

t is the **buyer's decision threshold**, and it is determined entirely by the buyer's own payoff asymmetry. This is the pivot on which §4.4 turns: t is private to the buyer, and no protocol parameter can observe it.

Table 3: Notation for §4, extending the table in §3.1.

Symbol	Meaning
V	buyer's gain from acting on a correct decision
K	buyer's loss from acting on an incorrect one
$t = K/(V + K)$	buyer's decision threshold
ϱ	price of one decision
g	per-attestation verification cost (gas)
G	screening value: welfare gain from an informative report
$\varphi\ B\ d$	flat bond, slashed fraction × bond × dispute rate (HeLa)
F	distribution of p over the seller's decision population

4.2 Plain x402: the report is noise

Proposition 3 (Uninformative reporting under plain x402). *In the decision purchase game with no penalty attached to the report, every $c \in [0,1]$ yields the seller the same payoff, and reporting $c = 1$ weakly dominates. The buyer's posterior is therefore independent of c, and it acts iff $E_F[p] \geq t$.*

Proof. The seller's revenue is ϱ if a sale occurs and 0 otherwise, and the report enters no other term of its payoff. A buyer that conditions on c purchases more often at higher c, so sale probability is weakly increasing in c and $c = 1$ is weakly dominant regardless of p. Because the seller's report is independent of p in equilibrium, c carries no information about p, and by Bayes' rule the buyer's posterior equals its prior $E_F[p]$ for every observed c. The buyer's action is therefore constant in c. ◻

This is not a criticism of x402, which is a payment protocol and does not claim to make claims trustworthy. It is a statement of what a payment rail leaves open, and it is the same gap the empirical ERC-8004 study found on the reputation layer: values that are "not commensurable" and "rarely grounded in verifiable interactions" [24]. Costless signals are uninformative signals.

The consequence is that the buyer must act on the unconditional mean. With our calibration population — Beta(5,2), mean $p = 0.7142$, chosen so the mean sits at the prototype's measured test accuracy of 0.715 (§6.1) rather than to flatter the result — and a buyer with $V = 100$, $K = 400$ (the 4:1 asymmetry the UCI dataset itself ships), the threshold is $t = 0.80$. Since $0.7142 < 0.80$ the buyer should abstain always; if it acts always, as an agent transacting at volume on an unconditional prior will, welfare is **−42.91** per decision.

4.3 The Brier regime, and the condition under which it does not pay

Attaching Definition 1 changes the seller's problem. By Proposition 1 the expected-loss-minimising report is $c = p$, so c becomes a sufficient statistic for the seller's private belief and the buyer can act iff $c \geq t$.

Proposition 4 (Participation condition for confidence attestation). *Let $G = E_F[max(0,\ pV - (1-p)K)] - E_F[pV - (1-p)K]$ be the screening value. Attaching a Brier attestation at per-decision cost g raises total welfare above plain x402* ***iff*** $\boldsymbol{g < G}$*, and the welfare gain is exactly $G - g$.*

Proof. Under Proposition 1 the seller reports $c = p$ and the buyer acts iff $p \geq t$, so per-decision welfare is $E_F[max(0,\ pV - (1-p)K)] - g$: the max arises because $pV - (1-p)K \geq 0$ exactly when $p \geq t$. Under Proposition 3 the report is uninformative and welfare is $E_F[pV - (1-p)K]$. Subtracting gives $G - g$, which is positive iff $g < G$. □

G is the value of being able to decline. It is bounded — a mechanism that costs more than the mistakes it prevents is not worth running — and this is where the measured gas figures bite.

Table 4: The participation condition at the measured per-attestation costs of §7.7. Welfare is per decision, in the buyer's units. $G = 56.19$.

Setting	Gas price	g	Welfare	Gain over x402	Verdict
Ethereum L1, busy	30 gwei	\$79.86	−66.58	−23.67	**does not pay**
Ethereum L1, quiet	10 gwei	\$26.62	−13.34	+29.57	pays
L2, typical	0.05 gwei	\$0.13	+13.15	+56.06	pays
L2, cheap	0.01 gwei	\$0.03	+13.25	+56.16	pays

On a busy L1 the mechanism is welfare-negative. The attestation costs more than the screening it buys, and a buyer is better off with an unattested decision and the gas in its pocket. This is §7.7's L2 conclusion arriving as a formal condition rather than a cost observation, and it sharpens it: the relevant comparison is not gas against the *price* of the decision but gas against G, the value of the errors the attestation lets the buyer avoid.

The condition also says when Brier is pointless for reasons having nothing to do with cost. $G \to 0$ when V and K are near-symmetric and F is concentrated above t: if almost every decision is worth acting on, screening buys nothing. **Confidence attestation is worth paying for exactly where decisions are consequential, asymmetric, and genuinely uncertain** — which is a much narrower market than "all agentic commerce," and we do not claim otherwise.

4.4 Against the alternatives, in the same model

§2.5 identified HeLa's accountability bond as the only true substitute in the agentic-payments literature. Modelling it here rather than describing it is what makes the comparison a result instead of a positioning claim.

Under a flat bond, a fraction φ of the bond B is slashed on an adverse verdict at dispute rate d. The expected penalty for selling a decision of quality p is $\varphi Bd(1 - p)$ — and, decisively, **it does not depend on *c***. The reporting margin is untouched, so Proposition 3 still applies and $c = 1$ remains optimal. What the bond constrains is the *participation* margin: the seller sells only when $\varrho > \varphi Bd(1 - p)$, that is when

$$p \geq 1 - (\varrho)/(\varphi\ B\ d).$$

The bond therefore implements a threshold too. It just implements it on the seller's side, by refusing to sell bad decisions, rather than on the buyer's side by reporting honestly.

Proposition 5 (Equivalence against a single buyer). *For any single buyer with threshold t, there exists a bond level $\varphi\ B\ d = \varrho/(1 - t)$ at which the flat-fraction mechanism achieves exactly the same welfare as the Brier mechanism, gross of attestation cost.*

Proof. The bond induces the seller-side threshold $p_{\min} = 1 - \varrho/(\varphi\ B\ d)$. Setting $\varphi\ B\ d = \varrho/(1 - t)$ gives $p_{\min} = t$. Both mechanisms then result in exactly the set $\{p \geq t\}$ of decisions being acted on, so both yield $E_F[max(0,\ pV - (1 - p)K)]$. □

The numbers confirm it: at $\varphi Bd = 100$ the bond attains 13.2757, and Brier attains 13.2757. **Identical, not merely close.**

This is a result that cuts against the mechanism and it should be stated plainly: *for a single buyer, confidence elicitation buys nothing a correctly tuned bond does not.* What Brier buys is robustness to the tuning being wrong, which is not nothing —

Table 5: Welfare when the bond is mis-set. The optimum is sharp in both directions.

Bond level	Induced threshold	Welfare	vs Brier
0.5 × optimum	0.60	−6.23	−19.51
optimum	**0.80**	**13.28**	**0.00**
2 × optimum	0.90	7.70	−5.58

— but robustness to mis-tuning is a weaker claim than the one §2.5 licenses, and on its own it would not justify a proving system.

4.5 Where the separation actually is: heterogeneous buyers

The equivalence in Proposition 5 quietly assumes one buyer, hence one t. Drop that and the mechanisms come apart, because they differ in **who** applies the threshold.

A bond acts on the seller, who sells or does not, once, for everyone. The threshold is baked into a protocol parameter and is necessarily the same for every buyer. A Brier attestation acts on the *report*: the seller publishes $c = p$, and each buyer applies its own t to the same number.

Proposition 6 (Pooling loss of threshold mechanisms). *With buyers heterogeneous in t, the Brier mechanism attains* $E_j\ E_F[\max(0,\ pV_j-(1-p)K_j)]$*, while any flat-bond mechanism attains at most* $\max_q\ E_j\ E_F[\mathbb{1}\{p \geq q\}(pV_j-(1-p)K_j)]$*. The first weakly dominates, strictly whenever two buyers have distinct thresholds.*

Proof. The Brier report $c = p$ is a sufficient statistic for p, so each buyer j solves its own problem and attains its first-best set $\{p \geq t_j\}$. A bond fixes one q for all buyers, so buyer j acts on $\{p \geq q\}$ rather than $\{p \geq t_j\}$. Where $t_j \neq q$ this set is wrong for j in one of two ways — it acts on decisions with $p < t_j$, or abstains on decisions with $p \geq t_j$ — and either strictly lowers j's payoff. Summing over buyers gives the result, with strictness whenever some $t_j \neq q$, which is unavoidable when thresholds differ. □

With four buyers whose downside risk escalates — a thin-margin lender, a standard one, a regulated one, a systemically exposed one, giving thresholds 0.50, 0.80, 0.90, 0.95:

Table 6: Heterogeneous buyers. The bond is tuned to its own optimum; it still cannot serve four thresholds with one parameter.

Mechanism	Threshold applied	Welfare
Brier	each buyer's own (0.50 / 0.80 / 0.90 / 0.95)	**15.83**
Flat bond, optimally tuned	0.89, pooled	4.67
	efficiency loss	**11.16 (70.5%)**

An optimally tuned bond loses **70.5%** of the available gain. The loss is not mis-tuning — the bond is at its optimum — it is the structural cost of compressing a continuous signal into one binary sell/don't-sell rule.

This is the mechanism's actual economic claim, and it is narrower than "accountability." Brier is not a better penalty than HeLa's. It transmits a different quantity: a bond communicates *the seller declined to sell you bad decisions at the protocol's threshold*, and an attestation communicates *here is how good this decision is; apply your own threshold*. The second is more valuable exactly when buyers differ, and worth nothing when they do not.

Two limits of the claim, stated here rather than in §7 because they bear directly on the result:

- **It assumes truthful adjudication** (Assumption 1). Proposition 1 supplies $c = p$, and Proposition 1 rests on the committee — tier 3. A colluding committee breaks §4.3 and §4.5 together.
- **It is a comparative-statics result, not a market prediction.** F, V, K and the buyer mix are assumptions, not measurements, and no agent market has been observed to behave this way. What the model establishes is a *conditional* claim: if buyers are heterogeneous in the way described, pooling costs this much. Whether real x402 buyers are heterogeneous is unmeasured.

4.6 What the model says about deployment

Three implications, none of which requires a new experiment:

1. **L2 or nothing.** §7.7 argued this on cost; Proposition 4 makes it a participation condition. Anything above roughly G per attestation makes the mechanism value-destroying rather than merely expensive.
2. **Target markets with heterogeneous, asymmetric buyers.** Credit, insurance underwriting, and clinical triage have exactly the property §4.5 needs. Uniform, low-stakes, high-volume

calls — the median x402 transaction today — have none of it.

3. **Do not claim to beat bonds in general.** Proposition 5 says a tuned bond matches Brier for a single buyer. Any comparison that omits it is overclaiming, and a reviewer who has read HeLa's whitepaper will find the omission in under ten minutes.

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5. System design

5.1 Components and where the circuit sits

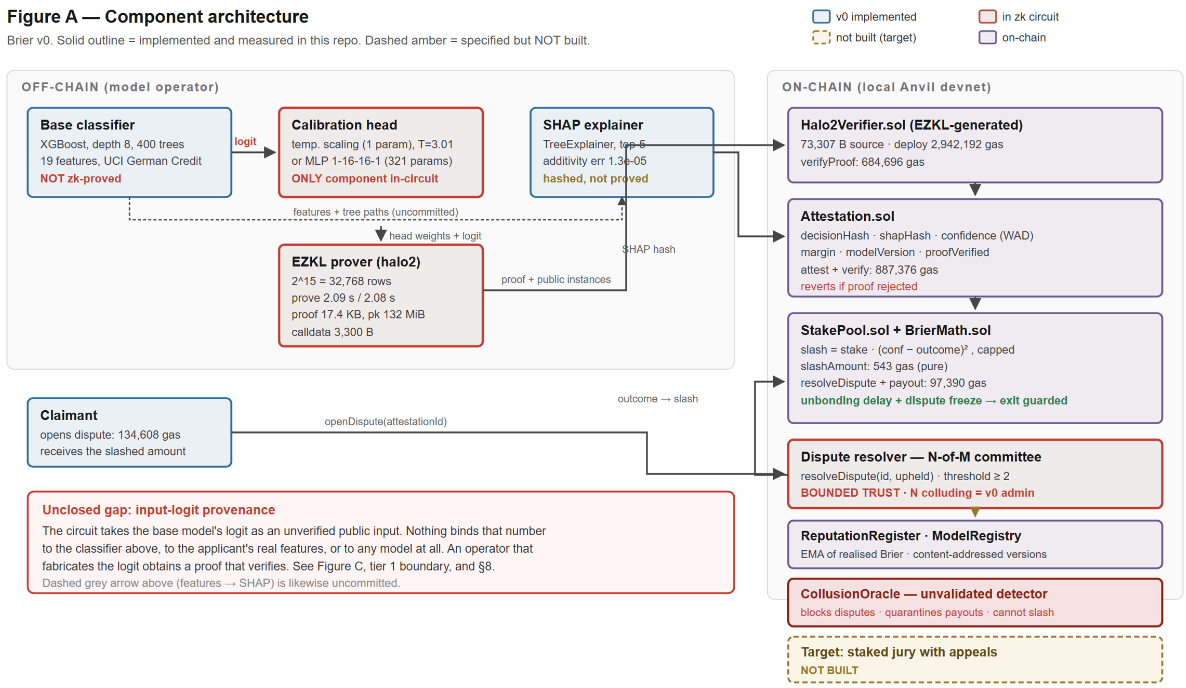

Figure 1: Component architecture. Solid outlines are implemented and measured; dashed amber is specified but unbuilt; red is the single component inside the zk circuit.

The base classifier and the SHAP explainer sit outside the circuit by design: the vector they produce is hash-committed as evidence, which establishes that the operator recorded a particular explanation and nothing about whether that explanation is faithful.

5.2 Protocol sequence and per-step cost

The sequence below is annotated with the gas each step actually cost on a local chain, so the expensive steps are visible rather than asserted: verification dominates, and the Brier arithmetic that is the substance of the mechanism is three orders of magnitude cheaper than proving that it ran. The diagram also marks the one step the mechanism still cannot enforce against: resolution is a committee action, so a penalty the contract computes correctly can be waived by the people who resolve it (§8.2).

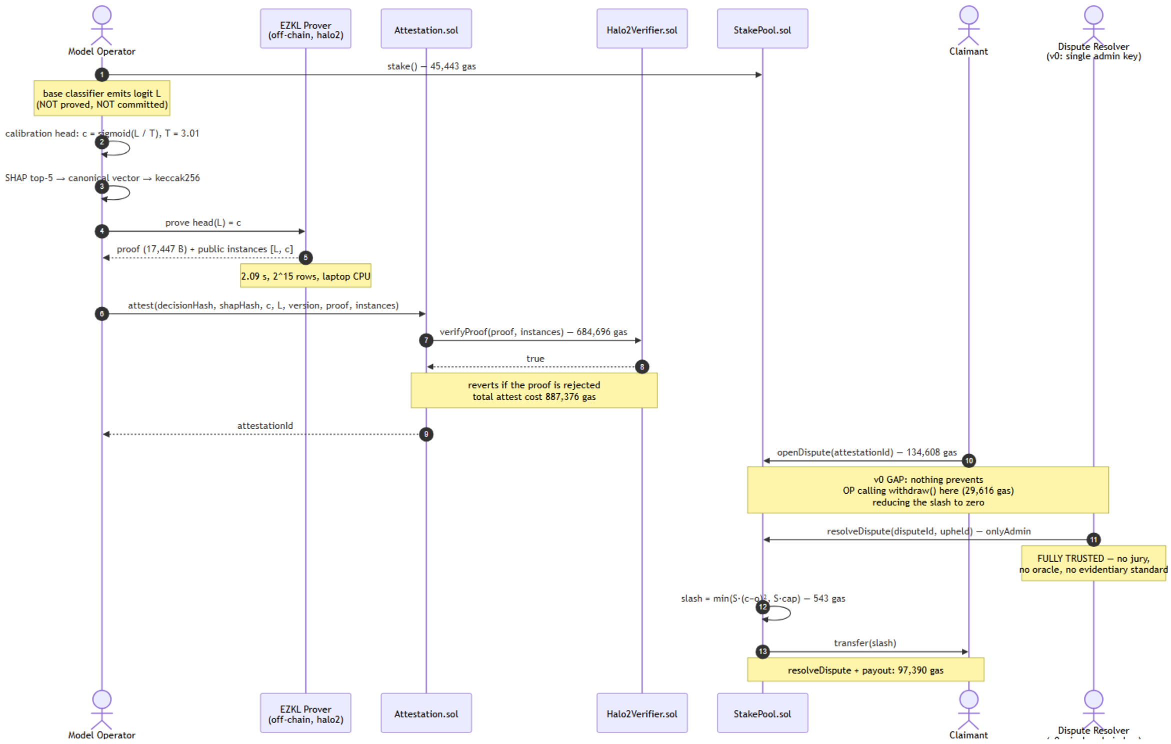

Figure 2: Protocol sequence, annotated with measured gas per step and the two points at which v0 is unenforced.

5.3 Trust boundaries

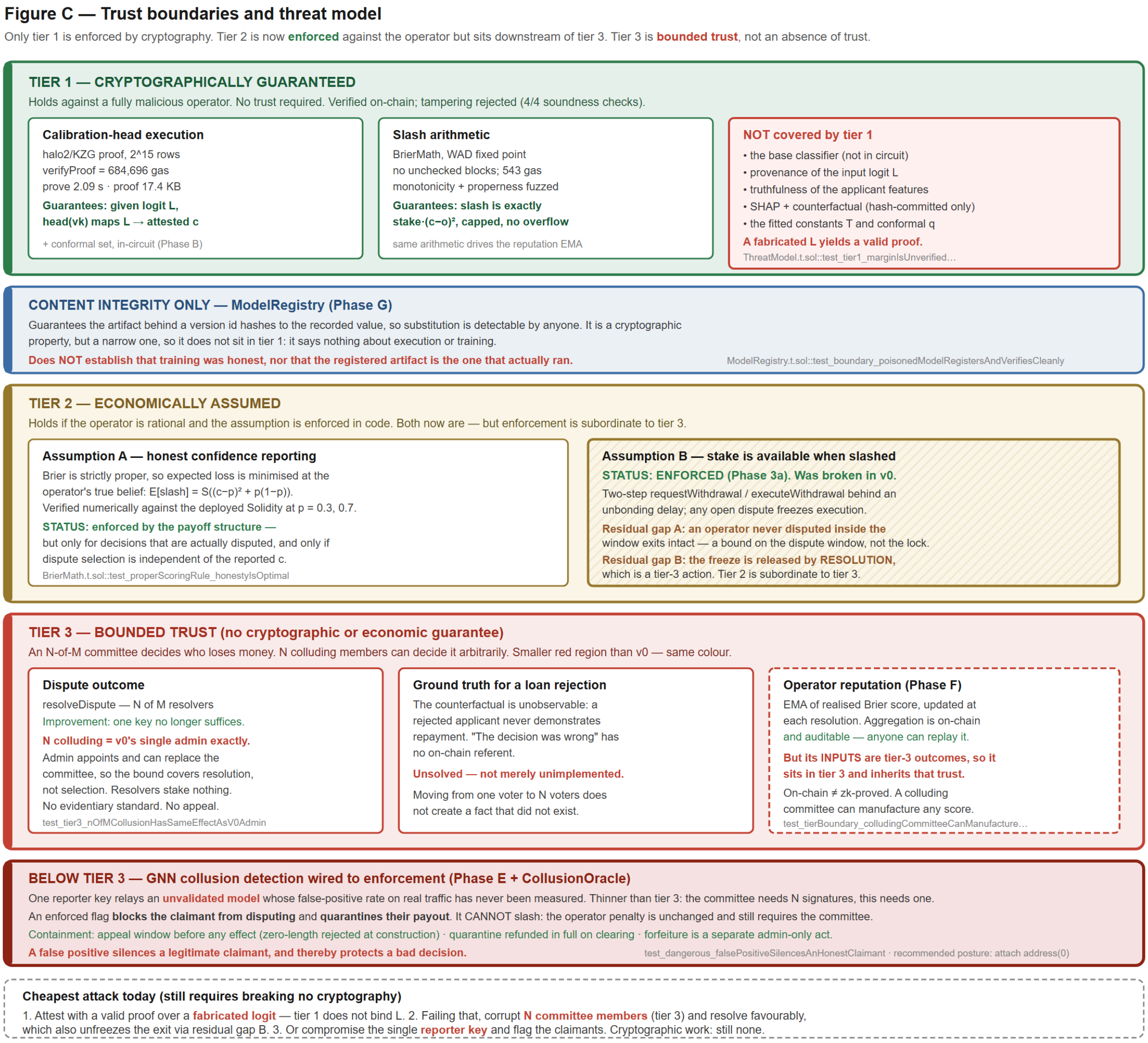

Figure 3: Trust boundaries. Green is cryptographically guaranteed, amber economically assumed, red fully trusted; each weakness names the test that demonstrates it.

This is the diagram against which every other claim in this document should be checked. Three tiers:

- **Tier 1 (cryptographically guaranteed, green).** Calibration-head execution and the slash arithmetic. Holds against a fully malicious operator.
- **Tier 2 (economically assumed, amber).** Assumption A (honest confidence reporting) is enforced by the payoff structure, conditionally. Assumption B (stake is available when slashed) was **broken in v0** and is now enforced: withdrawal is a two-step request/execute with an unbonding delay, and any open dispute freezes execution (§8.3). Two residual gaps remain, and neither is a timelock bug.
- **Tier 3 (bounded trust, red).** Dispute outcome and ground truth. An admin-appointed N-of-M committee decides who loses money; N colluding members can decide it arbitrarily. The tier keeps its colour — it is a smaller red region, not a different one.

Each weakness in the figure names the test that demonstrates it. The cheapest attack path still requires no cryptographic work: fabricate the logit, since tier 1 does not bind it (`test_tier1_marginIsUnverifiedOperatorSuppliedInput`), or corrupt N committee members, which `test_tier3_nOfMCollusionHasSameEffectAsV0Admin` shows buys exactly the power v0's single key had.

5.4 Calibration reliability

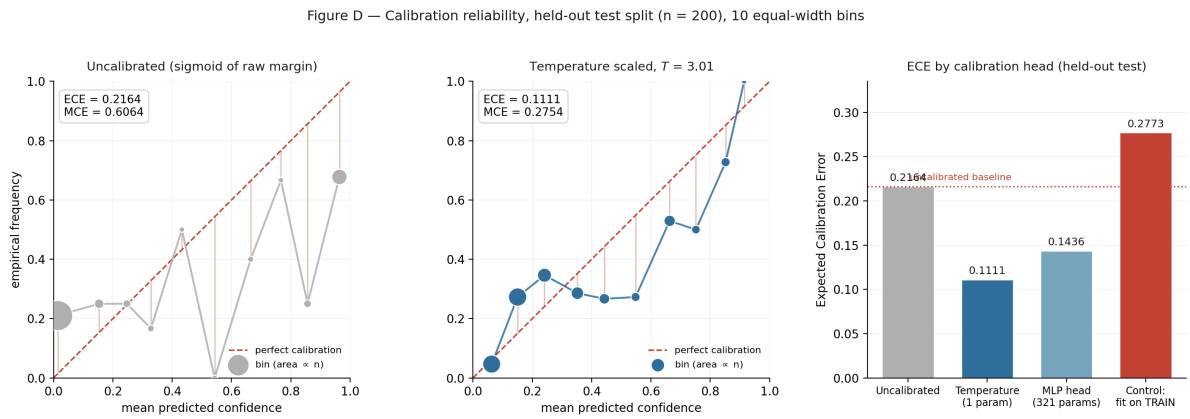

Figure 4: Reliability before and after temperature scaling, with the train-fitted leakage control at right. Bin values are read directly from the run artifacts; no curve is fitted.

Bin values are read directly from `artifacts/calibration/phase1_report.json`; no curve is fitted or smoothed. Left: uncalibrated, ECE 0.2164, with a bin at mean confidence 0.5430 whose empirical frequency is 0.0000. Centre: after temperature scaling with $T = 3.01$, ECE 0.1111. Right: ECE across heads, including the leakage control described in §6.2.

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6. Methodology

6.1 Dataset and model

UCI Statlog German Credit: 1,000 real credit applications, 20 attributes, downloaded at build time. The label is inverted so that the positive class is *reject*. Attribute 9 (personal status and sex) is dropped, leaving 19 features (see §8.4). Split 60/20/20 into train (600) / calibration (200) / test (200), stratified, seed 42, with mutual disjointness asserted in code.

The base model is XGBoost (depth 8, 400 trees, learning rate 0.3, no regularisation) and is **deliberately overfit**: train accuracy 1.0000, test accuracy 0.7150. This is a methodological choice, not a tuned result. Demonstrating a calibration-insurance mechanism on an already-calibrated model would establish nothing; the overfit regime reproduces the overconfidence that makes calibration worth insuring. The base model's accuracy is not a claim of this work.

6.2 Calibration

Two heads were implemented and both are proved:

Table 7: The two calibration heads. Both emit a logit, so the circuit stays affine plus ReLU and contains no transcendental function.

Head	**Parameters**	**Form**
Temperature scaling	1	$\sigma(L/T)$, T learned in log-space
MLP	321	$1 \rightarrow 16 \rightarrow 16 \rightarrow 1$, ReLU

Both emit a **logit**, with the sigmoid applied outside the module, so the circuit is affine + ReLU only and contains no transcendental function.

Both are fitted on the **calibration split only**. To show this matters rather than assert it, every run includes a control that fits the same head on the *training* split: that control yields $T = 0.1344$ and test ECE 0.2773, which is **worse than not calibrating at all** (0.2164). Fitting on training data *sharpens* an already-overconfident model.

6.3 Uncertainty quantification beyond a point estimate

Temperature scaling returns a calibrated scalar with no guarantee attached. Two stronger constructions were implemented and evaluated against it.

Split conformal prediction [16, 17] returns a *set* carrying a distribution-free marginal coverage guarantee: over the joint draw of calibration and test data, the true label lies in the set at least $1-\alpha$ of the time, with no assumption that the model is correct or even well specified. The calibration split is halved so the temperature and the conformal quantile never see the same points — reusing one split for both would break the exchangeability the guarantee rests on. Results in §A.1.

Deep-ensemble epistemic uncertainty [18] trains N independently seeded copies of the neural base model and feeds the spread of their predictions to the calibration head alongside the margin, on the intuition that a confident score the members disagree about should be discounted. Results in §A.2, where that intuition does not survive measurement.

6.4 Explainability

SHAP [19] TreeExplainer over the base model, top-5 attributions per decision, in margin space. Correctness is checked by additivity (SHAP values plus base value must reconstruct the model margin; measured max absolute error 1.26×10^{-5}) and by bit-identical reproduction across reruns. Five directional claims, written before inspecting output, are verified by correlating each feature's value against its attribution toward reject.

Counterfactual explanations. SHAP answers "what drove this decision"; an applicant reading an adverse action notice is asking "what would change it". Those are different questions, and the second is the actionable one. A greedy coordinate search over *actionable* features produces a minimal-ish change set that flips a rejection, restricted to values the training data actually contains. Immutable attributes — age, foreign-worker status, dependants — are excluded by construction and the exclusion is asserted rather than assumed. The counterfactual is hash-committed alongside the SHAP vector, in the same evidence slot and at the same trust tier: it is proof the operator recorded this explanation, not proof the explanation is faithful. Results in §A.3.

6.5 Evaluation metrics

ECE is the headline metric and the quantity the mechanism's value rests on, so its definition is given explicitly rather than by reference — the binning convention is exactly what a reviewer needs to audit, and different conventions give materially different numbers [13].

Definition 3 (Expected Calibration Error). Partition $[0,1]$ into B equal-width bins. For predictions $\{p\}_i$ with outcomes y_i, let I_b index the predictions falling in bin b. Then $ECE = \sum_b (|I_b|)/(n) \mid acc(I_b) - conf(I_b) \mid$, where $conf(I_b)$ is the mean prediction in the bin and $acc(I_b)$ the empirical frequency of $y = 1$ within it.

Three conventions are fixed, each of which changes the number:

1. $B = 10$, equal width, bins left-open and right-closed except the first, which includes 0.
2. Bins are weighted by population, so a bin holding two points cannot move the result as much as one holding two hundred.
3. **Empty bins are skipped, not counted as perfectly calibrated.** Counting them as zero-error would reward a model that concentrates its predictions into a narrow range, which is the opposite of what the metric should do.

Maximum Calibration Error is the same quantity under a max rather than a weighted mean, and is reported alongside because it exposes a single badly miscalibrated bin that ECE averages away. The Brier score needs no binning and is the quantity the mechanism actually charges, which makes it the least arbitrary of the three; it is decomposable into calibration and refinement terms, and only the first is what this mechanism prices.

The implementation is known-answer tested against hand-computed values (`tests/test_metrics.py`), because a silently wrong ECE would invalidate every calibration claim in §7 without failing anything.

6.6 Systems measurements

Beyond calibration: proof generation time (best of three runs, to suppress scheduler noise), proof size in bytes, calldata per attestation, proving-key size, native verification time, and on-chain verification gas. Circuit shape is recorded as `logrows` together with rows actually used, since the two diverge and only the first determines cost (§7.3).

6.7 Slashing validation

Monotonicity in miscalibration is verified deterministically at 101 confidence points and by fuzzing (256 runs per direction). Properness is verified numerically by scanning all 101 candidate reports at two distinct true probabilities, $p = 0.30$ and $p = 0.70$, and confirming the expected-loss minimiser equals p in both cases. Four scenarios are then run end-to-end on a local chain (§7.4).

---

7. Results

All figures below are produced by scripts in the repository. Environment: Windows 11, Python 3.13.5, xgboost 3.1.2, torch 2.9.1+cpu, EZKL 23.0.5, Foundry 1.7.1, laptop CPU, no GPU. Seed 42.

7.1 Calibration

The calibration claims rest on the 10-seed pass. Seeds are pinned in `config.EVAL_SEEDS` and the full list is always run, so a favourable subset cannot be reported selectively.

Table 8: Calibration across the 10 pinned seeds. Temperature scaling reduces ECE in every seed; the train-fitted control is worse than not calibrating in every seed.

Head	ECE mean ± std	Brier mean ± std	Accuracy mean ± std
Uncalibrated	0.1853 ± 0.0248	0.2060 ± 0.0191	0.7505 ± 0.0209
Temperature (1 param)	**0.0870 ± 0.0218**	**0.1758 ± 0.0113**	0.7505 ± 0.0209
MLP (321 params)	0.1055 ± 0.0234	0.1908 ± 0.0119	0.7330 ± 0.0199
Control: fitted on TRAIN	*0.2434 ± 0.0212*	—	—

Learned $T = 3.4657 \pm 0.4491$, with $T > 1$ in **every** seed — the base model required softening everywhere, never sharpening. Temperature scaling reduces ECE in **every** seed, and the train-fitted control is worse than not calibrating at all in **every** seed (paired Wilcoxon $p = 0.002$, the minimum attainable at $n = 10$). Accuracy is unchanged by temperature scaling, as it must be: a monotone rescaling of the logit cannot move the decision boundary. Calibration buys a better-priced confidence, not a better classifier, and the table separates the two.

Negative result on head capacity, reported at the strength the data supports. The 321-parameter MLP head does not beat the 1-parameter head. Across the 10 pinned seeds temperature scaling has the lower ECE in **7 of 10** — paired Wilcoxon $W = 7.0$, $p = 0.037$, median difference -0.018. This is a *majority* result, not a uniform one, and it is stated that way deliberately: a single run per head would have supported the stronger and less accurate claim. With 200 calibration points the extra capacity usually fails to generalise, but in 3 of 10 seeds the MLP wins. The reported strength is pinned by `test_temperature_beats_mlp_claim_matches_reported_strength`, so overstating it later would fail the suite. The convenient part of this result is that the better estimator is also the cheaper circuit; that agreement is a measured outcome rather than a design assumption.

The result replicates on a second dataset. Everything above is one corpus of 1,000 rows, which is thin ground for the paper's central empirical claim. §8.6 runs the identical protocol — same splits, same seeds, same head, same fitting, same metrics, and deliberately the same base-learner hyperparameters — on UCI *Default of Credit Card Clients*: 30,000 rows from a Taiwanese bank, labelled with **observed defaults** rather than an analyst's credit grade.

Table 9: The same five claims, on both datasets. Means over the same 10 pinned seeds.

Claim	German Credit	Taiwan default
ECE reduced in every seed	10/10	10/10
Mean ECE reduction	52.8%	54.8%
Learned $T > 1$ (base model overconfident)	3.47 ± 0.45	2.80 ± 0.06
Realised Brier — the quantity slashed	0.2060 → 0.1758	0.1642 → 0.1490
Accuracy unchanged (monotone map)	yes	yes

The label change is the one that carries weight. A calibration finding that held on adjudicated opinion but not on realised outcomes would be a much weaker result, and these two datasets are the pair that can distinguish those.

The base model is under-fit on the larger corpus, since its hyperparameters were chosen to overfit 1,000 rows and were not retuned — retuning per dataset would have tested whether the data can be modelled well rather than whether the finding travels. Absolute accuracy is therefore not the comparison to read; the ECE reduction is, and it is what the mechanism prices.

Seed-selection audit. Because a single seed can be chosen after the fact, we report where ours falls. Seed 42 ranks 6th of 10 by ECE reduction (48.7% against a 52.8% mean) while carrying the *highest* uncalibrated ECE of any seed — the single-seed figures are therefore pessimistic in absolute terms and middling in relative terms. `SEED = 42` was committed once, before any result existed, and never changed; the repository contains no notebooks.

Seed 42 in detail, since it is the seed the figures, circuits and end-to-end run are generated from:

Table 10: Seed 42 in detail, the seed from which the figures, circuits and end-to-end run are generated.

Head	Params	ECE	MCE	Brier
Uncalibrated	—	0.2164	0.6064	0.2353
Temperature scaling	1	**0.1111**	0.2754	0.1897
MLP	321	0.1436	0.7766	0.2030
Control: temperature fitted on TRAIN	1	*0.2773*	—	—

Learned $T = 3.0121$; base accuracy train 1.0000, test 0.7150. Figure 4 plots the left two columns of this table as reliability curves.

7.2 Explainability

Additivity max absolute error 1.26×10^{-5}; attributions bit-identical across 3 reruns; 5/5 directional sanity checks pass (`installment_rate_pct_income` $+0.621$, `duration_months` $+0.505$, `credit_history` -0.756, `checking_status` -0.943, `savings_status` -0.834).

7.3 Proving and verification

Table 11: Proving and verification cost for the two heads. A 321x parameter increase changes neither the circuit size nor the proving time.

Metric	Temperature (1 param)	MLP (321 params)
Circuit size	$2^{15} = 32{,}768$ rows	$2^{15} = 32{,}768$ rows
Proving time	**2.09 s**	**2.08 s**
Native verification	0.094 s	0.058 s
Keygen	1.86 s	2.03 s
Proof size	17,447 B	17,566 B
Proving key	132 MiB	132 MiB
Calldata per proof	3,300 B	3,300 B
Tamper-soundness	4/4 rejected	4/4 rejected

Proving cost is invariant to a 16,897× parameter increase. A 10-point sweep from 1 to 16,897 parameters holds `logrows` = 15 at every point. Rows used scale linearly with parameter count (slope 0.98 rows/param, $r = 0.992$) but proving time does not follow it: mean 2.13 s, standard deviation 0.09, regression slope +0.009 s per decade of parameters, Spearman rank correlation -0.188 at $p = 0.603$ — indistinguishable from no relationship. Fixed lookup-argument and column overhead dominate the multiply-accumulate count entirely. This is the answer to RQ3, and it is the reason the design proves a head rather than a classifier: within the circuit's capacity, a larger head is very nearly free, so the binding constraint is capacity, not parameter count.

Scope limit. This holds *within* logrows = 15. The largest head fills 16,511 of 32,768 rows (50.4%), so the sweep does not cross the capacity boundary; where cost steps at logrows 16+ is **unmeasured**. The claim is "flat until the head stops fitting the circuit," not "flat without limit."

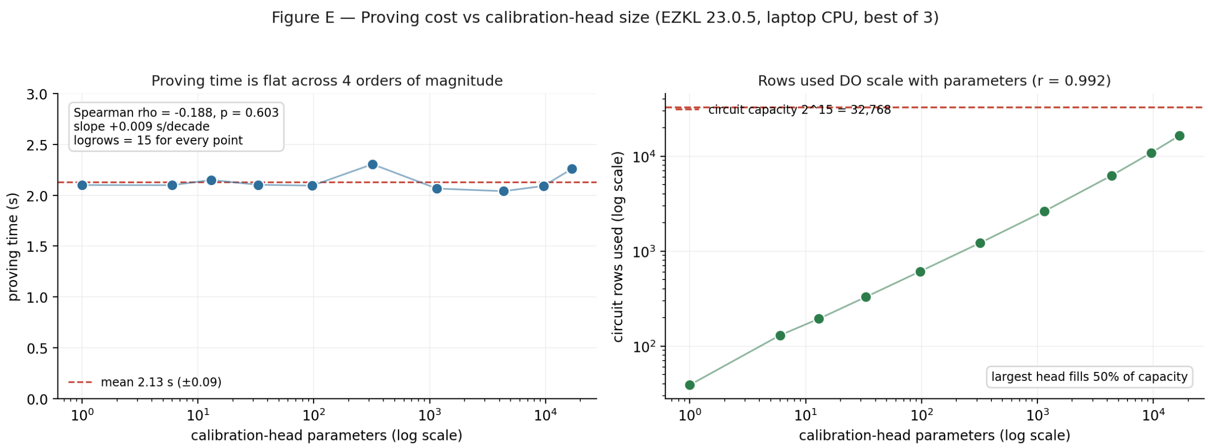

Figure 5: Proving cost against head size across four orders of magnitude. Rows used scale linearly with parameters; proving time does not follow.

Soundness is tested, not assumed: an honest proof verifies, while a tampered public output, a flipped byte at the proof head, a flipped byte mid-proof, and a wrong verifying key are each rejected.

7.4 On-chain cost and end-to-end behaviour

Table 12: Measured gas per operation. Verification dominates; the Brier arithmetic that is the substance of the mechanism is roughly 0.08% of the per-decision cost.

Operation	**Gas**
`Halo2Verifier` deployment (one-off)	2,942,192
`verifyProof` (real EZKL proof)	**684,696**
`attest` (verify + store)	887,376
`openDispute`	134,608
`resolveDispute` incl. slash + payout	97,390
`stake`	45,443
`withdraw`	29,616
`BrierMath.slashAmount` (pure)	543

Verification is ~33× a plain 21,000-gas transfer. The Brier arithmetic is 543 gas — approximately 0.08% of the per-decision cost. **Essentially all cost is halo2 verification, not the mechanism.**

End-to-end, on a local Anvil chain, with a real proof generated per decision:

Table 13: End-to-end behaviour on a local chain, one real zk proof generated per decision. Percentages are of the stake remaining as each scenario ran.

Scenario	Confidence	Outcome	Slash (% of stake)	Prove
(a) confident + correct	0.9845	upheld	**0.0239%**	2.24 s
(b) confident + wrong	0.9367	overturned	**87.7458%**	2.25 s
(c) uncertain + wrong	0.5012	overturned	**25.1173%**	2.06 s

Ordering holds: (b) > (c) > (a). Confident-and-wrong costs 3,667× confident-and-correct and 3.49× uncertain-and-wrong. Percentages are of the stake remaining when each scenario ran — the three execute sequentially against one shrinking stake, so the percentage column is the comparable one and absolute ETH values across rows are not.

These figures are now enforceable against the operator, which was not true in v0. The earlier exit path — front-running `openDispute()` with an instant `withdraw()` — reduced every slash in this table to zero. Withdrawal is now a two-step request/execute separated by an unbonding delay, and an open dispute freezes execution, so `ThreatModel.t.sol::test_tier2_frontRunDisputeByWithdrawing_isNowBlocked` asserts the attack fails and is the standing regression test for it. What the table still does not survive is a dishonest committee: resolution is a tier-3 action, so the figures describe what the mechanism computes and enforces against the *operator*, not what it enforces against its own resolvers (§8.2, §8.3).

7.5 Five enhancements, ablated against the baseline

Each candidate was run under the identical seed protocol as the baseline — same 10 pinned seeds, same splits, same calibration discipline — and kept only if it earned its place. Two did not, and one of those actively hurt. Full per-phase detail, including the capacity control that makes the ensemble result interpretable, is in **Appendix A**; `ABLATION.md` carries the generated tables.

Table 14: Ablation verdicts. "Kept" means the enhancement is in the shipped pipeline. Effects are over 10 pinned seeds against the same baseline.

Enhancement	Effect on the priced quantity	Circuit cost	Verdict
Conformal prediction (§A.3)	Distribution-free coverage, marginal	No measurable proving cost	**kept**
Deeper base model (§A.1)	Null — no calibration gain	—	rejected
Deep-ensemble epistemic uncertainty (§A.2)	**Degrades** calibration	—	rejected
Counterfactual evidence (§A.4)	Evidence only, not proved	Off-circuit	kept, tier 1
Collusion detection (§A.5)	Synthetic rings only; FPR unknown	Off-circuit	kept, **unvalidated**

Two of these results are worth stating in the body rather than leaving to an appendix. The deep-ensemble arm is the clearest: epistemic uncertainty is standard practice and it made the calibration *worse*, which is only interpretable because a capacity control was run alongside it — the architecture helps, the epistemic signal destroys the gain (§A.2). And the collusion detector is validated against synthetic rings only, so its false-positive rate on real traffic is unknown; §8.5 states what follows from attaching an unvalidated model to money.

7.6 Two on-chain surfaces, and where they actually sit

Both additions are on-chain and neither is a cryptographic guarantee. Being on-chain establishes that a computation is public and replayable; it says nothing about whether its inputs can be trusted, and the two are routinely conflated.

Operator reputation (Phase F). An exponential moving average of realised Brier score per operator, updated at each resolution from the same $(c,\ o)$ pair that drives the slash — so the score and the penalty cannot diverge. The aggregation is on-chain and auditable: anyone can replay the update rule against the emitted events. But its *inputs* are resolved dispute outcomes, which come from the committee. **The score therefore sits in tier 3 and inherits that trust level exactly**, which `test_tierBoundary_colludingCommitteeCanManufactureReputation` asserts by having two colluding resolvers destroy an honest operator's record with no recourse. What it adds is legibility, not assurance: a number computed by a rule fixed in advance, so an operator's history is visible without trusting anyone's summary of it.

The obvious gaming path is closed by the mechanism itself. An operator farming disputes on decisions it knows are correct still pays $(c-1)^2$ on every sample and cannot reach zero — the irreducible $p(1-p)$ of §3.2, appearing as contract behaviour. Wiring is optional: a pool constructed without a register slashes identically, so reputation can never become load-bearing for the penalty.

Model-version registry (Phase G). A halo2 verifying key covers a *program*, not an *input*: one key verifies every proof from that circuit whatever weights were compiled into it. So `model_version` cannot be inferred from a proof and must be an explicit commitment, or an attestation silently means "some version of this head ran". The registry records a content hash per version, making substitution detectable by anyone.

Its guarantee is content integrity and nothing more, and two tests exist purely to pin that boundary: a deliberately poisoned model registers and verifies perfectly cleanly (`test_boundary_poisonedModelRegistersAndVerifiesCleanly`), and the proof does not bind a version id to the circuit that ran (`test_boundary_anyoneMayReferenceAVersionTheyDidNotRun`). Training integrity is a different and much harder problem and is out of scope. If either test ever needs changing, the claim has been inflated.

7.7 What the mechanism costs an operator to participate in

§8.6 records that the staking parameters are demonstration values with no actuarial basis. They can be given one, and the pieces are already measured: the expected slash per dispute is the operator's realised Brier score (§3.2), and the per-attestation gas is §7.4. What follows is a parameter-selection framework rather than a recommendation, since it turns on a dispute rate this prototype cannot observe.

Expected loss is the Brier score, and calibration is what lowers it. By Proposition 1 a truthful operator's expected slash per disputed decision is $S \cdot p(1-p)$, whose empirical counterpart is the realised Brier score. For the calibrated head that is 0.1758 (§7.1): a well-calibrated operator loses about **17.6% of stake per dispute in expectation**, and no reporting strategy lowers it, because the term is irreducible.

The uncalibrated model's Brier is 0.2060. Calibration therefore reduces the operator's expected per-dispute loss by roughly 15%, which is the mechanism's answer to why anyone would calibrate: not compliance, but a smaller bill.

Table 15: Expected per-decision cost as a fraction of stake, at three dispute rates. Only the dispute rate is hypothetical; the Brier figures are measured over the 10 pinned seeds.

Dispute rate	Uncalibrated (B = 0.2060)	Calibrated (B = 0.1758)	Saving
0.1%	0.0206%	0.0176%	15%
1%	0.2060%	0.1758%	15%
5%	1.0300%	0.8790%	15%

Verification cost, and why an L2 is not optional. At 887,376 gas per attestation, the per-decision fee is a hard floor on the value of a decision worth attesting at all.

Table 16: Cost of one attestation, at 887,376 gas and ETH at $3,000. The mechanism's own arithmetic is 543 gas of that total; the rest is halo2 verification.

Setting	Gas price	Cost per attestation
Ethereum L1, busy	30 gwei	$79.86
Ethereum L1, quiet	10 gwei	$26.62
L2, typical	0.05 gwei	$0.13
L2, cheap	0.01 gwei	$0.03

At L1 prices the design is only coherent for decisions worth hundreds of dollars each, which excludes most consumer credit. At L2 prices it is negligible against any lending decision. **The practical claim here is therefore an L2 claim**, and §7.4's L1 gas figures are best read as an upper bound rather than a target. Proof aggregation across decisions would amortise verification further and is not implemented.

Capital efficiency is the real cost of the unbonding period. An operator locks S for the unbonding delay τ beyond the last disputable decision. At an opportunity cost r, the carrying cost is $r\ \tau\ S$ per cycle, and this is what W1a must trade against the dispute window: a longer τ closes residual gap A and raises the carrying cost linearly. At $\tau = 7$ days and $r = 5\%$ annually the carry is about 0.096% of stake per cycle — comparable to a single dispute at a 0.5% dispute rate, so at these parameters the timelock is not the dominant cost. That ceases to be true if τ must grow to months to cover loan-rejection claim latency, which is precisely the case §8.3 flags as unresolved.

Participation condition. Writing f for the fee per decision, n for decisions per cycle, d for the dispute rate and B for realised Brier, an operator participates when

$$f\ n > n\ d\ B\ S + r\ \tau\ S + n\ g$$

with g the per-attestation gas cost. Two of the four right-hand terms scale with stake and two do not, so the binding constraint changes with S: at small stake, gas dominates and the design is L2-or-nothing; at large stake, the $d\ B\ S$ term dominates and calibration quality becomes the operator's main lever. Estimating d empirically is the missing input, and it can only come from a deployment.

7.8 Test suite

144 Solidity tests (Foundry), 98 Python tests, and 11 middleware integration tests against a live chain — 253 in total, all passing.

Table 17: The Solidity suites. ThreatModel and CollusionOracle assert that documented weaknesses are reachable, not that the system is secure.

Suite	Tests	What it holds down
`BrierMath.t.sol`	23	Fixed-point properness, monotonicity, fuzzing, the 100% cap
`StakePool.t.sol`	24	Staking, disputes, slash cap, payout failure
`ModelRegistry.t.sol`	19	Tamper detection, and the boundary of that claim
`CollusionOracle.t.sol`	18	Quarantine, appeal window, and the reversibility of a flag
`ReputationRegister.t.sol`	16	EMA arithmetic and reputation-farming paths
`ERC8004ReputationAdapter.t.sol`	13	Sign inversion, and that a dead registry cannot halt resolution
`ThreatModel.t.sol`	11	The documented weaknesses, as executable evidence
`Unbonding.t.sol`	10	The withdrawal-freeze invariant, fuzzed over timing
`ReputationIntegration.t.sol`	6	Reputation driven by real resolutions
`Verifier.t.sol`	4	Real-proof verification and tamper rejection

The middleware suite runs against a real Anvil chain with the real contracts deployed rather than a mocked RPC, and one of its tests pins a *limitation* rather than a feature: the gate does not prevent an attestation being replayed across requests [25], and if a future change adds claim-once semantics that test should fail and be rewritten.

Two of these carry unusual weight. `ThreatModel.t.sol` does **not** assert the system is secure — it asserts that each documented weakness is real and reachable, so a failure there means the threat model has drifted from the code and this paper is wrong. And `tests/test_claim_vocabulary.py` greps this document for language that would attribute properties the system lacks (§8.2); the forbidden list lives in the test rather than in prose, so a later edit cannot quietly upgrade a claim.

8. Limitations and threat model

The subsections below state each unresolved gap in full. The table summarises which adversary capabilities the system covers and which it does not; the uncovered rows are part of the stated security model, not omissions from it.

Table 18: Threat coverage. "Covered" means an adversary with that capability is either prevented or reliably detected; every uncovered row is a capability an adversary retains today.

Adversary capability	Covered	Mechanism, or reason not
Forge a proof for a head it did not run	yes	halo2 soundness; 4/4 tamper checks
Alter an attested confidence after the fact	yes	stored on chain, proof-bound
Substitute model weights under a claimed version	yes	ModelRegistry content hash (§7.6)
Exit stake ahead of a pending dispute	yes	unbonding delay + dispute freeze (§8.3)
Act alone as a single resolver	yes	N-of-M threshold (§8.2)
Fabricate the input logit L	**no**	tier 1 binds the head, not the pipeline (§8.1)
Corrupt N of M resolvers	**no**	equals v0's single-key power (§8.2)
Exit before any dispute is raised	**no**	bound on the dispute window, not the lock (§8.3)
Misreport within a protected subgroup	**no**	calibration is measured in aggregate (§8.4)
Flag an honest claimant	**no**	detector unvalidated on real traffic (§8.5)
Train dishonestly and register truthfully	no	content integrity ≠ training integrity (§7.6)
Decline to attest at all	no	unattested decisions are outside the system

Five capabilities in bold are the ones an adversary would actually use. Each has a subsection below and a test that demonstrates it.

These are unresolved trust gaps in the present design. They are not future work.

8.1 Only the calibration head is proved

Theorem 1 (Scope of the tier-1 guarantee). *For an attestation accepted on chain with verifying key vk and public instances (L, c), the proof establishes that the calibration head identified by vk maps L to c. It establishes nothing about the provenance of L.*

The proof binds a computation, not a pipeline. The base classifier is not in any circuit. **An operator that fabricates L obtains a proof that verifies**, as demonstrated by `test_tier1_marginIsUnverifiedOperatorSuppliedInput`, in which the chain accepts `type(int256).max` as a margin and still marks the attestation proved. Any description of this system as "zero-knowledge proving the AI decision" is false.

A secondary point: the proof attests the **quantised** computation. At input/param scale 13, the in-circuit fixed-point head differs from the float head by up to 4.2×10^{-4}. The on-chain confidence is the quantised one.

8.2 Dispute resolution is bounded trust, not an absence of trust

`resolveDispute` now requires N of M signatures from a fixed resolver committee, replacing v0's single admin key. A single resolver cannot act alone (`test_tier3_singleResolverCannotResolveAlone`). That is the entire improvement, and it should be read narrowly: **the number of keys an adversary must corrupt goes from 1 to N, and nothing else about the trust model changes.**

Specifically, and each of these is asserted by a passing test:

- N colluding members have exactly the power the single admin had — they can slash a well-calibrated operator to zero (`test_tier3_committeeCanSlashAnHonestOperator`) or shield a miscalibrated one (`test_tier3_committeeCanShieldAMiscalibratedOperator`), which `test_tier3_nOfMCollusionHasSameEffectAsV0Admin` states as the headline case.

- The admin appoints and can replace the committee (`test_tier3_adminCanReplaceTheCommittee`), so the bound applies to the *resolution* step, not to committee *selection*.
- Resolvers have nothing at stake. A resolver who votes dishonestly loses nothing, so there is no economic mechanism holding the tier honest — which is why it is not tier 2.
- There is still no evidentiary standard. Resolvers vote on a bare boolean, and nothing in the contract defines what evidence to consult or penalises consulting none.

`docs/PHASE3B_TRUST_MODEL.md` fixes this vocabulary, and it was written before the code it describes so the claim could not drift upward once the tests went green.

Beneath the implementation gap sits a harder problem, and moving from one voter to N voters does not touch it: for a loan *rejection* the counterfactual is unobservable, because a rejected applicant never demonstrates repayment. "The decision was wrong" has no on-chain referent. This is unsolved, not merely unimplemented.

Beneath the implementation gap sits a harder problem: for a loan *rejection* the counterfactual is unobservable, because a rejected applicant never demonstrates repayment. "The decision was wrong" has no on-chain referent. This is unsolved, not merely unimplemented.

8.3 The unbonding period closes the exit path, and two gaps remain

Proposition 7 (Withdrawal cannot outrun an open dispute). *Let withdrawal require a request followed, after an unbonding delay $\tau > 0$, by a separate execution, and let execution revert while the operator has any open dispute. Then no sequence of operator actions completes a withdrawal between the opening and the resolution of a dispute against it.*

Proof. Execution is gated on the open-dispute count being zero, and that count is incremented by `openDispute` before any subsequent block. An operator whose dispute is open therefore fails the gate at execution time regardless of when the request was made, and a request made after the dispute opens must in addition wait τ, during which the dispute remains open. □

The floor on the delay is one hour, enforced at construction, with seven days used in tests. `Unbonding.t.sol` fuzzes the invariant over randomised interleavings of request, dispute and execution rather than testing one sequence, and `test_tier2_frontRunDisputeByWithdrawing_isNowBlocked` is the standing regression for the specific v0 exploit — if it ever passes again, tier 2 has regressed.

Two residual gaps survive, and neither is fixed by a longer timelock:

Gap A — exit before any dispute is raised. An operator that requests withdrawal and is not disputed within the unbonding window exits cleanly, and a later dispute lands against an operator with nothing at stake (`test_tier2_residualGap_exitBeforeAnyDisputeIsRaised`). This is a bound on the *dispute window*, not a timelock bug: the unbonding period must exceed the time a claimant realistically needs to notice a bad decision and act.

That parameter now has a derivation, and it makes the tradeoff worse than v0 admitted. `docs/UNBONDING_PERIOD_JUSTIFICATION.md` derives it from the US statutory chain governing algorithmic credit decisions: 60 days for the consumer's free-file window (15 U.S.C. § 1681j, cited in § 1681m(a)), 30 days for the statutory reinvestigation (§ 1681i(a)(1)(A)), and 15 days for the permitted extension (§ 1681i(a)(1)(B)) — $\boldsymbol{\tau}$ = **105 days**, fifteen times the value used in the tests. Each term is a statutory maximum, which is the right posture for a security parameter, since the operator chooses the timing.

Re-running §7.7's carrying cost `carry = r · τ · S` at r = 5% inverts that section's conclusion.

Table 19: Carrying cost against the expected slash per cycle (realised Brier 0.1758, §7.1). The 7-day row is the one §7.7 characterised as "comparable to a single dispute"; the 105-day row is what the statute implies.

τ	Carry per cycle	Expected slash at 1% dispute rate	Carry dominates below
7 days	0.0959%	0.1758%	0.545% dispute rate
105 days	**1.4384%**	0.1758%	**8.18% dispute rate**

At a defensible τ, capital lockup is the **dominant** cost across the entire plausible range of dispute rates — roughly 8× the expected slash at 1%. Three consequences follow, and the first is the most damaging:

- **It weakens the incentive argument of §7.7.** That section argues calibration pays because it cuts expected loss by 15%. If the slash is an eighth of the total cost, that saving is about 1.7% of what participation actually costs.
- **It selects for cheap capital.** Carry and slash are both linear in stake, so the ratio is scale-invariant; access to a low r is not.
- **It is not a tuning problem.** τ is pinned by statute above and by capital cost below, and the two do not meet. Closing it needs a different instrument — bonded insurance, rolling tranche release, or slashing from an at-risk fraction smaller than the full stake. None is implemented, and none is a parameter change.

The derivation is US- and credit-specific, and its own scope conditions are stated where it lives: notice delivery has no statutory deadline, so the clock starts at notice rather than at decision; other regimes and non-credit decisions yield different numbers. `MIN_UNBONDING_PERIOD` is unchanged and the tests still use seven days, which exercise mechanism behaviour rather than parameter choice. What changed is that a deployer now has an arithmetic to accept rather than a guess to inherit.

Gap B — tier 2 is enforced downstream of tier 3. The freeze is released by dispute *resolution*, which is a committee action. A dishonest committee can resolve favourably to unfreeze an operator's exit (`test_tier2_residualGap_adminCanUnfreezeByResolvingFavourably`). Tier 2 is therefore enforced against the operator but remains subordinate to tier 3, and closing it requires progress on §8.2 rather than on the timelock.

8.4 Fairness

Attribute 9 (personal status and sex) is excluded from the feature set. This is a floor, not a fairness guarantee: proxy discrimination through correlated features is untested, no disparate-impact analysis was performed, and — most relevant to this mechanism — calibration is measured in aggregate and **not within protected subgroups**. A model can be well calibrated overall while badly miscalibrated for a subgroup, which is precisely the harm this mechanism should detect and currently does not.

The conformal layer of §A.3 does **not** fix this, and it would be easy to imply otherwise. Its coverage guarantee is *marginal*: a conformal predictor can hit 90% coverage overall while covering a protected subgroup at 60%. A stronger uncertainty claim about the population average is still a claim about the population average.

A measurement was attempted, and it produced a null result that closes off the obvious fix. The plan was the one §9.4 sets out: construct a model that is aggregate-calibrated but subgroup-miscalibrated, show aggregate ECE misses it, implement a within-group variant that catches it, and track that variant on-chain. `scripts/21_subgroup_adversary.py` runs it against attribute 9, the excluded protected attribute — the natural adversarial target, since the model cannot see it while an operator holding the applicant records can condition on it freely.

The first-order result looked like a clean confirmation: within-group ECE exceeds aggregate ECE in 10 of 10 pinned seeds, mean 0.1349 against 0.0870, a 55% understatement. It does not survive a control. Permuting the subgroup labels while holding group sizes fixed reproduces the entire effect — a worst-group gap of 0.0559 for random partitions against 0.0548 for the real one, Wilcoxon signed-rank $p = 0.92$, with the real partition beating its own null in 3 of 10 seeds.

The cause is a small-sample bias in ECE itself. Within-bin accuracy is estimated from a handful of points and |acc − conf| is an absolute value, so sampling error accumulates instead of cancelling.

Table 20: ECE of a model calibrated *by construction* (p ~ Uniform(0,1), y ~ Bernoulli(p)), so the entire reported value is the estimator's own bias. 10 bins, 400 draws per size.

n	68	106	200	500	2,000	10,000
ECE	0.1188	0.0989	0.0695	0.0451	0.0223	0.0098

The test split is 200 rows and the two measurable subgroups hold 68 and 106. **At n = 68 the noise floor (0.1188) exceeds this pipeline's aggregate ECE (0.0870)** — the estimator's bias at subgroup sizes is larger than the quantity being measured.

Three consequences, and none of them is that the mechanism is subgroup-fair:

1. **Nothing here shows the mechanism is subgroup-calibrated.** Absence of measurable evidence is not evidence of absence.
2. **The obvious fix is invalid at this scale.** A within-group ECE gate on 1,000 rows would fire on noise, and slashing an operator for it would be slashing them for an artifact of the auditor's estimator. The on-chain subgroup register that would have tracked it was **not built**, because building it would have made the gap look closed.
3. **It reframes what is needed.** Roughly 2,000 rows per subgroup puts the floor near 0.02. That is a dataset ~20× this one, or a debiased estimator with a characterised noise floor — equal-mass binning, or a kernel estimator with a bootstrap interval. Both are in §9.4.

Resolved on a larger dataset, and the effect is real but small. Point 3 named the missing ingredient as roughly 20× the data, and §8.6's replication supplies it: the Taiwan default dataset has subgroups of 2,336–3,664 test rows against German Credit's 54–116. Run there, the identical analysis — same binning, same protected-attribute construction, same permutation control — gives a worst-group gap of 0.0067 against a permuted null of 0.0042, beating its null in 8 of 10 seeds at $p = 0.037$.

So the question was answerable all along; this dataset simply could not answer it. Subgroup miscalibration **is** present, and it is roughly 12% of the aggregate ECE — real, and much smaller than the 55% understatement the naive German Credit comparison appeared to show before the null was run. Both halves of that matter: the effect exists, and the first measurement of it was noise.

This does not license a subgroup slashing rule. An enforcement gate needs an estimator whose bias is characterised at the group sizes it will encounter, and binned ECE at n = 68 has a bias of 0.1188 — twenty times the effect just measured. A rule that fires on a real 0.0067 in a regime where the estimator manufactures 0.1188 is not measuring what it claims to. W4a in §9.4 is unchanged, and `SubgroupReputationRegister.sol` still does not exist.

`tests/test_subgroup_calibration.py` pins the null against erosion: the permutation control, the minimum group size, and the noise floor's monotonicity in n are all asserted, so a later change that makes the gap "significant" by shrinking a threshold fails loudly rather than quietly.

8.5 An unvalidated detector has been given authority over money

This is the sharpest limitation in the system, and unlike the others it was chosen rather than inherited.

§A.5 reports detection performance on rings that were injected, so the labels exist by construction. There is no labelled real collusion, the protocol having never run, and the false-positive rate on genuine dispute traffic is therefore unknown. An earlier draft of this document said the detector was not wired to any enforcement action and must not be until that number exists. It is now wired, and for real traffic the number still does not exist.

What can be measured has now been measured, and it is unfavourable. The false-positive rate on real traffic is unobtainable, but a weaker and still decision-relevant quantity is not: on synthetic traffic containing *no rings at all*, every flag is false by construction, so the rate at which the detector fires there is a true false-positive rate. `scripts/24_detector_fpr.py` measures it at the threshold the contract actually enforces — `CollusionOracle.MIN_SCORE = 0.8e18`, below which `flag()` reverts — training on one dispute graph and scoring a disjoint second one.

Table 21: Detector behaviour at the enforced threshold of 0.80, over the 10 pinned seeds. FDR is the fraction of flagged claimants who were honest.

Traffic	FPR	Recall	FDR
No rings present	**0.94%** (95% Wilson upper bound **1.51%**)	—	100% by construction
Rings at intensity 0.85	2.41%	87.2%	**19.9%**
Rings at intensity 0.60	1.67%	48.9%	**23.5%**
Rings at intensity 0.40	1.79%	16.1%	**50.0%**

Three things follow, and none of them is reassuring.

The detector flags honest claimants in a world with no collusion in it. 17 of 1,800, across ten ring-free graphs. That rate is small but it is not zero, and each instance is a legitimate claimant silenced for the appeal window.

One flagged claimant in five is honest, at the easiest setting. The FDR is the number a flagged individual actually faces, and it is far worse than the FPR because ring members are rare. At intensity 0.40 the detector recalls 16% of ring members while half of everyone it flags is innocent — a coin flip that silences bystanders.

§A.5's headline AUC of 0.998 does not describe this. AUC is threshold-free and integrates over operating points the contract cannot use. The deployed system has one operating point, fixed at 0.80, and its precision there is the number that governs whether attaching the oracle is defensible. Reporting AUC and enforcing at a threshold is how a system comes to look validated while being unvalidated for what it does.

This does not close the limitation. It is synthetic traffic from a generator whose realism is itself an assumption, and a detector that behaves this way on ring-free synthetic traffic has cleared the lowest available bar, not a meaningful one. What changes is that an operator deciding whether to pass a non-zero oracle address now has a measured number where there was none, and a reproducible procedure they can re-run on their own traffic once they have any. On this evidence the honest recommendation is to pass `address(0)`.

What an enforced flag does: blocks the claimant from opening disputes, and diverts any payout they win into quarantine. What it does not do: slash. The operator's penalty is computed and taken identically whether or not the claimant is flagged, because a suspected colluder does not make a bad decision retroactively correct — and because letting a flag reduce a slash would hand operators a motive to get their own claimants flagged.

Four properties bound the damage, each asserted by a test:

1. A flag never causes a slash.
2. An appeal window precedes any effect; a zero-length window is rejected at construction, so a false positive cannot bite in the block it was raised.
3. Quarantined funds are reversible in full when a flag is cleared.
4. Permanent confiscation is a separate, admin-only act, so the only path from a model output to an irreversible loss runs through a person.

None of that makes the detector correct. `CollusionOracle.t.sol` carries three `test_dangerous_*` cases which exist to keep this legible: a false positive silences an honest claimant and thereby *protects* a genuinely bad decision; the reporter key can flag anyone, with nothing on-chain distinguishing a model output from a lie; and flag-plus-forfeit confiscates a legitimate award with recourse only to the admin that took it.

In trust terms this sits below tier 3. The committee at least requires N signatures. This is one key relaying an unvalidated model. Deployments that have not measured the detector's false-positive rate on their own traffic should attach no oracle at all, which `test_poolWithoutOracleIgnoresFlagsEntirely` shows costs nothing else.

8.6 Scope of the empirical claims

The calibration results now rest on **two** datasets, and the second was chosen to differ on the axes that could have been driving the first. UCI *Default of Credit Card Clients* (Taiwan, 2005) is 30,000 rows against German Credit's 1,000, from a different country and decade, and — the difference that matters most — its label is an **observed default event** rather than an analyst's credit grade. A calibration finding that held only on adjudicated opinion and not on realised outcomes would be a much weaker finding, and these two datasets are the pair that can tell those apart.

The protocol was held fixed rather than retuned: the same split fractions, the same 10 pinned seeds, the same head, the same fitting procedure and the same metric definitions, imported from the modules the first pipeline uses. The base learner's hyperparameters were deliberately *not* adjusted, even though they were chosen to overfit 1,000 rows and are wrong for 30,000. Retuning them would have answered a different question. The consequence is that the base model is underfit here, so absolute accuracy is not the comparison to read.

Table 22: Replication on a second dataset. Every row is a claim that could have failed. Figures are means over the same 10 pinned seeds.

Claim	German Credit	Taiwan default	Holds
Temperature scaling reduces ECE in every seed	10/10	10/10	yes
Mean ECE reduction	52.8%	**54.8%**	yes
Learned temperature exceeds 1 (base model overconfident)	3.47 ± 0.45, 10/10	2.80 ± 0.06, 10/10	yes
Calibration lowers realised Brier — what the slash prices	0.2060 → 0.1758	0.1642 → 0.1490	yes
Accuracy unchanged (monotone rescaling)	identical	identical	yes

The ECE reduction is significant at $p = 0.002$ (Wilcoxon signed-rank over seeds). The core empirical claim of this paper therefore travels: the base model is overconfident and a one-parameter head fixes roughly half of that, on two datasets that share almost nothing but their domain.

A gap this dataset closes that the first could not. §8.4 reports that the subgroup-calibration question was unanswerable on German Credit, because binned ECE's small-sample bias at n = 68 exceeded the effect being measured. Taiwan's subgroups hold 2,336–3,664 test rows, well past the noise floor, and the same permutation control that killed the first result is passed here: the worst-group gap is 0.0067 against a permuted null of 0.0042, beating its null in 8 of 10 seeds at $p = 0.037$.

So **subgroup miscalibration is real** — and the finding cuts both ways. It vindicates §8.4's refusal to call the first null a clean bill of health, and it bounds the effect: 0.0067 of ECE is small in absolute terms, roughly 12% of the aggregate. A subgroup slashing rule would still need an estimator whose bias is characterised at the group sizes it will actually see, which §9.4's W4a describes and this work does not provide.

What remains outside the evidence: two datasets are still both credit, both tabular, and both binary — one model family and one decision class. Calibration drift over time is not evaluated on either. Staking parameters remain demonstration values with no actuarial basis, and correlated tail risk — one bad model version producing thousands of simultaneous claims — is not modelled.

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9. Open problems

§8 is a list of reasons the present system should not be trusted. This section states what would remove each of them, ordered by what blocks what. Every item names the evidence that would count as having closed it, so these are falsifiable targets rather than intentions — and two items below are recorded as closed by measurements reported in this paper, one of which closed unfavourably and opened a harder problem in its place.

9.1 Enforcement: shipped, with the parameter question still open

W1 — Unbonding period. Done (Phase 3a). Withdrawal is a two-step request/execute separated by an unbonding delay, and an open dispute freezes execution. The acceptance criterion was that `test_tier2_operatorCanFrontRunDisputeByWithdrawing` invert; it now reads `..._isNowBlocked` and asserts the attack fails, with `Unbonding.t.sol` fuzzing the invariant across randomised interleavings rather than one sequence (§8.3).

W1a — Choosing the unbonding period honestly. Open. Residual gap A shows the timelock only helps if it exceeds the time a claimant needs to notice a bad decision and act. For loan rejections the counterfactual may take months, so seven days is almost certainly too short.

W1a — Done, and the answer is unfavourable. `docs/UNBONDING_PERIOD_JUSTIFICATION.md` derives $\tau = 105$ days from the FCRA chain (§8.3) and re-runs the capital-efficiency arithmetic with it. The deliverable asked for "the real design tension, still unquantified"; it is now quantified, and it is worse than the phrasing anticipated. Carry becomes the dominant cost below an 8.18% dispute rate, roughly 8× the expected slash at 1%.

What remains open is not the number but the instrument: τ is bounded below by statute and above by capital cost, and those bounds do not meet, so no choice of τ is good. *Deliverable, revised:* a mechanism that decouples the dispute window from the locked-capital window — bonded insurance, rolling tranche release, or an at-risk fraction below the full stake — with the same treatment §7.7 gives the current design. This is a new work item, not a closed one.

9.2 Making the attested confidence mean something

W2 — Input-logit provenance. Three paths, in decreasing order of guarantee strength and increasing order of practicality:

Table 23: Candidate routes to binding the input logit, in decreasing order of guarantee strength.

Path	Guarantee	Cost	Status
Prove the base classifier in-circuit	Cryptographic, end-to-end	Prohibitive for depth-8, 400 trees	Unmeasured; §7.3 suggests capacity, not parameters, is the wall
Commit to feature vector and model weights	Fabrication becomes *detectable*, not *impossible*	Cheap — hashes only	Closest to shippable
Attest the base model in a secure enclave	Hardware trust replaces cryptographic trust	Cheap	Strictly the weakest of the three

The honest ordering matters: only the third is cheap today, and it is the one that gives up the most. *Evidence:* the first path is answered by measuring where proving cost steps past logrows 15 for a real ensemble — the boundary §7.3 explicitly did not cross. That measurement is the immediate next experiment.

W3 — Adjudication beyond bounded trust. Partly shipped (Phase 3b), and the remainder is the hard part. The single admin key is gone: resolution now needs N of M committee signatures, so an adversary must corrupt N keys rather than one (§8.2). That is the whole of the improvement, and `test_tier3_nOfMCollusionHasSameEffectAsV0Admin` exists to stop it being read as more. What remains is everything that would make the tier *accountable* rather than merely wider: a Kleros-style juror pool drawn from a staked set [7], an evidentiary standard, appeals, and resolvers with something at risk. *Evidence:* `test_tier3_committeeCanSlashAnHonestOperator` and `test_tier3_committeeCanShieldAMiscalibratedOperator` must both become unreachable, and `test_tier2_residualGap_adminCanUnfreezeByResolvingFavourably` with them, since gap B is downstream of this tier.

W3 has a hard research problem underneath it, and it should not be presented as an integration task. For a loan *rejection* the counterfactual is unobservable — a rejected applicant never demonstrates repayment — so "the decision was wrong" has no on-chain referent. Any juror pool needs a definition of the adjudicated event before it can vote on one. Candidate framings worth evaluating: restrict disputes to decisions with an observable counterfactual (approvals that defaulted); adjudicate *process* compliance rather than outcome; or score against a delayed, independently sourced outcome oracle. None is obviously right.

9.3 Strengthening the empirical claims

W3b — Bind the committed constants. §A.3 promoted conformal set construction into the circuit, but the quantile q joins T as a fitted constant committed off-chain. Every strengthening of the *proved* computation so far has left the *inputs* to that computation unproven, which is the same shape as the tier-1 gap in W2. Committing *(T, q)* through the Phase G registry and binding the registry entry into the circuit's public instances would close the constant half of it, and is cheap. It does not touch the logit half.

W4 — Subgroup calibration as the slashed quantity. §8.4 is the sharpest scientific gap: a model can be well calibrated in aggregate and badly miscalibrated for a protected subgroup, which is precisely the harm this mechanism should price and currently cannot see. Slashing on within-group ECE rather than aggregate confidence would target it directly. *Evidence:* a demonstration that the current rule fails to penalise a model constructed to be subgroup-miscalibrated but aggregate-calibrated, and that the revised rule does.

W5 — External validity. Partly done (§8.6). The calibration claims now rest on two datasets rather than one: the Taiwan default corpus replicates all five core claims under a fixed protocol, and its size resolved the subgroup question §8.4 could not answer. *Still open:* a second **model family** — both datasets run the same XGBoost base learner, so nothing here separates "this calibration result holds generally" from "this holds for gradient-boosted trees" — and a calibration-drift study over time, which remains unevaluated on either dataset.

W6 — Validate the detector that is already enforcing. Measured, and the measurement argues against it (§8.5). The graph analysis recovers injected rings (§A.5) and now has an enforcement path, which inverts the usual ordering: the capability shipped before the evidence that would justify it.

The false-positive rate at the enforced threshold has now been measured on synthetic ring-free traffic — 0.94%, 95% Wilson upper bound 1.51% — along with a false *discovery* rate of 19.9% at the easiest ring intensity and 50% at the hardest. That is enough to settle the deployment question in the negative without settling the validation question at all: **the honest posture is to pass `address(0)`**, and it now rests on a number rather than on the absence of one.

Still required, in order of urgency: a labelled corpus of **real** disputes, without which no false-positive rate on real traffic is obtainable and this limitation cannot close; a precision target agreed in advance, since the current threshold was chosen before any precision figure existed; and an appeal reviewer who is not the admin able to forfeit.

W7 — Why did the ensemble not help? §A.1 found that ensembling improves uncalibrated ECE and that calibration then absorbs the gain entirely, and §A.2 found that an epistemic-uncertainty signal actively harms calibration on 200 calibration points. Both suggest the binding constraint is the *size of the calibration split*, not model capacity. That is a testable claim: sweep the calibration split size and see whether the two-input head's advantage appears once it has enough data. If it does, the negative results of §A.1–§A.2 are statements about this dataset rather than about the approach, and they should be restated as such.

9.4 Sequencing

The items above are not independent, and the ordering below reflects what unblocks what. W2 is placed first because every other guarantee in the system is conditional on the input it does not currently bind.

Table 24: Proposed sequencing. "Answers" names the question each item closes; "Blocked by" names what must land first.

#	Item	Answers	Blocked by	Status
1	W2a — measure where proving cost steps past logrows 15	Is in-circuit provenance affordable at all?	—	open
2	W2b — commit feature vector and weights	Makes fabrication detectable	1	open
3	W1a — set the unbonding period from claim latency	How long is the dispute window?	—	**done, §8.3**
3b	W1b — decouple dispute window from capital lockup	Can a defensible τ be afforded?	3	**new, opened by 3**
4	W6 — measure detector false-positive rate	Should the oracle be attached?	—	**measured, §8.5 — answer is no**
5	W4 — subgroup calibration as the slashed quantity	Does the rule price the harm it should?	5b	**blocked, §8.4**
5b	W4a — a debiased calibration estimator, or 20× the data	Can subgroup ECE be measured at all?	—	**half done: at 20× the data the effect is real (§8.6); the estimator is still uncharacterised**
6	W3 — staked jury, evidentiary standard, appeals	Can tier 3 be made accountable?	3	open
7	W5 — second dataset, second model family, drift	Do the empirical claims travel?	—	**dataset done, §8.6; family and drift open**

Four rows changed shape rather than simply closing, and every change came from attempting the work rather than from re-reading the plan. That is the pattern worth noting: each item that was actually executed came back different from how it was specified.

W1a is done and spawned W1b. Deriving τ answered the question and revealed that no value of τ is satisfactory (§8.3). The follow-on is a different instrument, not a better parameter.

W6 is measured and the answer is negative. The false-positive rate at the enforced threshold now exists (§8.5), and it settles the *deployment* question — pass `address(0)` — without settling the *validation* question, which still needs real labelled disputes. An item can be answered without being closed.

W5 is half done, and the remaining half is the harder one. A second dataset replicated every core claim (§8.6). But both datasets run the same base learner, so nothing yet separates "this calibration result holds generally" from "this holds for gradient-boosted trees" — and a second model family is a cheaper experiment than the one already done, so its absence is now the more glaring gap.

W4 stays blocked, but W4a is half answered. §8.4's null was about the estimator rather than the mechanism, and the larger dataset shows the effect is real once group sizes permit measuring it (§8.6). What is still missing is the part that would license enforcement: an estimator whose bias is characterised at the group sizes a deployment would actually see. Binned ECE's bias at $n = 68$ is twenty times the effect measured at $n = 2{,}336$, so a gate built on it would fire on the estimator, not the model. `SubgroupReputationRegister.sol` still does not exist.

Items 1, 5b and the open half of 7 are independent and can proceed in parallel. Item 6 remains the largest and still depends on a definition of the adjudicated event that does not yet exist (§8.2).

9.5 Explicitly out of scope

Non-deterministic decision classes (LLM outputs). We flag this as **substantially harder, not a natural extension.** The present design depends on a deterministic decision with a scalar confidence and a binary outcome. Free-form generation has no canonical scalar confidence, no binary ground truth, and no stable notion of "the same decision" across sampling seeds. Applying a proper scoring rule there requires first defining the event space being scored, which is an open research problem and not an engineering task. We would rather name it as out of scope than imply the mechanism generalises for free.

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10. Conclusion

Returning to the six questions of §1.3.

RQ1 is answered affirmatively. The Brier slash has a unique expected-loss minimum at the operator's true subjective probability (§3.2), and the deployed fixed-point Solidity reproduces that minimum numerically at two distinct true probabilities, with monotonicity fuzzed in both directions.

RQ2 is answered affirmatively, with the cost stated plainly. The calibration head proves in 2.09 s on a laptop CPU and verifies on-chain for 684,696 gas — roughly 33× a plain transfer. The mechanism's own arithmetic is 543 gas, about 0.08% of the per-decision cost. Essentially all cost is halo2 verification, not confidence-weighted slashing.

RQ3 is answered, within a stated boundary. Proving cost is flat across a 16,897× parameter increase; capacity, not parameter count, is the binding constraint. The boundary itself — where cost steps past logrows 15 — is unmeasured, and measuring it is the next experiment.

RQ5 is answered negatively, and usefully so. Of five enhancements ablated against the baseline, conformal prediction earns its place — a genuinely stronger uncertainty claim, in-circuit at no measurable cost — while a deeper base model is null and deep-ensemble epistemic uncertainty is actively harmful. The capacity control in §A.2 is what makes the second result interpretable: the architecture helps, the signal destroys the gain. The one-parameter head of §7.1 survives every attempt made here to beat it, which is a stronger endorsement of it than the original single-run result was.

RQ6 is answered conditionally, and the condition is restrictive. Attaching a confidence attestation to a priced decision raises welfare only where its verification cost falls below the screening value it buys (Proposition 4); at the L1 gas prices measured in §7.4 it does not, so the practical claim is an L2 claim by necessity rather than preference. Against a single buyer an optimally tuned flat-fraction bond achieves *exactly* the same welfare (Proposition 5), so confidence elicitation buys nothing there that a correctly set parameter does not. The separation requires buyer heterogeneity, where a bond compresses quality into one pooled threshold while an attested confidence is a sufficient statistic each buyer applies its own threshold to — worth 70.5% of the available gain across the buyer mix we model (Proposition 6). That heterogeneity is modelled, not measured, and the claim is conditional on it.

RQ4 is answered negatively, and that is the substantive finding. A proved calibration head is *not* sufficient to make an attestation trustworthy. The proof binds the map from logit to confidence and nothing else: an operator that fabricates the logit obtains a proof that verifies, and the chain accepts it. Beyond that, dispute resolution is an admin-appointed N-of-M committee whose collusion carries exactly the authority a single key did, and while the unbonding period now stops an operator outrunning a dispute, an operator that is simply never disputed inside the window still exits with the stake intact. Each of these is demonstrated by a passing test, not conceded in prose.

The two on-chain surfaces added since do not change this answer, and were built so as not to appear to. Reputation is auditable aggregation over tier-3 inputs; the registry gives content integrity, not training integrity. Placing each in its real tier was the work, not adding them.

The economic question is narrower than the mechanism-design one, and §4 answers it against the mechanism more often than for it. Attaching a confidence attestation to an x402-priced decision is worth its verification cost only where that cost falls below the value of the errors screening prevents; at the L1 gas prices measured in §7.4 it does not, so the practical claim is an L2 claim by necessity rather than by preference. More pointedly, against a single buyer an optimally tuned flat-fraction bond — the mechanism HeLa already deploys — achieves *exactly* the same welfare, so confidence elicitation buys nothing there that a correctly set parameter does not. The separation is real but specific: a bond compresses quality into one pooled sell/don't-sell threshold, while an attested confidence is a sufficient statistic each buyer applies its own threshold to, and with buyers spanning thresholds of 0.50 to 0.95 an optimally tuned bond still forfeits 70.5% of the available gain. The mechanism's economic case rests on buyer heterogeneity, which this work models and does not measure.

Three further measurements came back against the design, and are reported here at the same weight as the results that favour it.

Deriving the unbonding period from statute rather than convenience gives 105 days, at which capital lockup becomes an operator's dominant cost — roughly 8× the expected slash — which substantially weakens §7.7's argument that calibration pays for itself.

The subgroup-calibration gap that §8.4 names as the sharpest scientific gap turned out to be unmeasurable on the first dataset: the apparent effect is entirely reproduced by a random partition, because ECE's small-sample bias at $n = 68$ exceeds the quantity being measured. On a 30,000-row dataset the same analysis resolves it, and the effect is real but small — about 12% of aggregate ECE. Both halves matter. The effect exists, *and* the first measurement of it was noise; a paper that had reported only the first would have overstated the problem sevenfold (an apparent 0.0479 against a measured 0.0067), and one that reported only the second would have missed that the instrument was the thing being measured. The on-chain register was not built either way, because binned ECE's bias at realistic group sizes is twenty times the effect it would be enforcing on.

The third is about a component that had already shipped. The collusion detector reports an AUC of 0.998 in §A.5 and is wired to withhold a claimant's payout. At the single threshold `CollusionOracle.sol` actually enforces, it flags 0.94% of honest claimants on traffic containing no collusion at all, and at the easiest ring intensity one flagged claimant in five is honest. Nothing about the detector changed; what changed is that it was evaluated at the operating point the contract uses rather than integrated over operating points the contract cannot reach. **Reporting a threshold-free metric while enforcing at a fixed threshold is how a system comes to look validated for something it is not**, and the recommendation is now explicit: pass `address(0)`.

A negative result that blocks a planned deliverable is still a result, and burying any of these would have left the repository implying gaps had closed that had not.

The trajectory is worth stating precisely, because it is the argument for the approach. Two of v0's three named gaps have closed since the first draft, and neither closure required weakening a claim: the exploit tests were *inverted*, and the vocabulary guard in `tests/test_claim_vocabulary.py` prevents the resulting language from drifting upward. What did not move is tier 1 — the input logit is unverified today for exactly the reason it was then — and the ground truth problem beneath tier 3, which more signatures cannot touch.

That invariance is the most durable result here. A better base model, a stronger uncertainty quantifier, a richer evidence bundle, a reputation history and a version registry all now sit on top of the same unproven input. Sophistication above the trust boundary does not move the trust boundary, and this work is a fairly thorough demonstration of that.

The two claims should therefore be separated cleanly. The **mechanism-design claim** — that confidence-weighted slashing on a strictly proper scoring rule makes honest confidence reporting loss-minimising, and that it is cheap to implement and verify — is supported by the evidence presented here. The **systems claim** — that this constitutes a trust-minimised accountability protocol — is not supported by it. §9 states what would close that distance, distinguishing the items this work closed from those it did not, and records that one of the closures made the problem harder rather than easier. The most useful result here may be the negative one: it establishes precisely which parts of "verifiable AI accountability" the cryptography actually buys, and which parts remain a matter of whom you trust.

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Appendix A — Per-phase ablation detail

Five candidate enhancements, each measured against the shipped baseline on the identical seed protocol. §7.5 gives the verdicts; this appendix gives the evidence behind each, including the two that failed.

A.1 A deeper base model does not help

An entity-embedding neural network over the same 19 features, and its ensemble with XGBoost, were evaluated on the identical 10-seed protocol. The XGBoost arm reproduces §7.1 exactly, which is the check that the protocol is the same one rather than merely a similar one.

Table 25: Phase A. The ensemble improves uncalibrated ECE and temperature scaling then absorbs the difference entirely.

Model	Uncalibrated ECE	Calibrated ECE	Calibrated Brier	Accuracy
XGBoost (baseline)	0.1853 ± 0.0235	0.0870 ± 0.0207	0.1758 ± 0.0107	0.7505 ± 0.0198
Tabular NN	0.2419 ± 0.0189	0.0874 ± 0.0226	0.1843 ± 0.0071	0.7280 ± 0.0187
Ensemble	0.1617 ± 0.0181	0.0834 ± 0.0154	0.1733 ± 0.0091	0.7410 ± 0.0164

Null result. On calibrated ECE the ensemble wins 6 of 10 seeds ($p = 0.6953$) and the NN alone 5 of 10 ($p = 0.9219$) — neither distinguishable from chance. On accuracy the ensemble *loses*, 2 of 10 ($p = 0.1953$). It is not adopted.

The informative part is where the gain goes. The ensemble clearly improves *uncalibrated* ECE (0.1617 against 0.1853), and temperature scaling then absorbs the whole difference. **Ensembling and calibration are substitutes here, not complements.** Since the ensemble costs an entire second model family and calibration costs one parameter, this is an argument for the one-parameter head rather than against sophistication in general.

A.2 Epistemic uncertainty makes calibration worse

5 independently seeded network copies; disagreement is the standard deviation of member probabilities, supplied to the head alongside the margin.

Table 26: Phase C. The capacity control is the informative row: the same head fed a constant beats the baseline, so the epistemic signal is what destroys the gain.

Arm	ECE	Brier	Accuracy
Temperature (baseline, 1 param)	0.0870 ± 0.0207	0.1758 ± 0.0107	0.7505 ± 0.0198
Margin + disagreement	0.1072 ± 0.0324	0.1916 ± 0.0174	0.7295 ± 0.0294
Margin + constant (capacity control)	0.0743 ± 0.0170	0.1757 ± 0.0109	0.7530 ± 0.0165

Negative result, and the control is what makes it legible. The epistemic signal *degrades* calibration: 2W/8L against the baseline, median +0.01611, $p = 0.1309$. But the same two-input head shape fed a *constant zero* in place of the signal beats the baseline — 8W/2L, median -0.01238, $p = 0.0371$.

So the architectural change helps slightly and the epistemic signal then destroys that gain. Without the control this would have read as "ensemble disagreement doesn't help"; with it, the finding is sharper and less comfortable — the signal is **noise that the head overfits** on 200 calibration points. The cost is not close either: 5 model copies at 54 s per seed, roughly ten times the baseline, to make calibration worse.

The wider lesson is about experimental design more than about ensembles. Had the control been omitted, the arms would have read as "epistemic uncertainty does not help", and the true effect — that the architecture helps while the signal cancels it — would have been invisible. Any ablation that adds capacity and a feature simultaneously needs the capacity-only arm to be interpretable.

A.3 Conformal prediction, and its circuit cost

Empirical coverage across the pinned seeds, at three levels:

Table 27: Phase B. Coverage tracks the target and sits 1-2 points below it at every level, within sampling noise at n = 10.

Target	Empirical coverage	Min over seeds	Seeds ≥ target	Avg set size
0.80	0.7930 ± 0.0399	0.7300	5/10	1.102
0.90	0.8815 ± 0.0408	0.8200	4/10	1.357
0.95	0.9335 ± 0.0398	0.8550	4/10	1.570

Coverage tracks the target and sits 1–2 points below it at every level. At $n = 10$ the shortfall is within sampling noise, so this is **not** reported as a violated guarantee — nor as confirmation. The likely cause is that the three-way split is stratified, which mildly breaks the exchangeability conformal assumes; on genuinely iid synthetic draws the implementation does meet its target (`tests/test_conformal.py`). Set size is reported alongside coverage because coverage alone is trivially satisfiable: always return both labels and you have 100% coverage and no information.

The circuit gate, measured, not argued.

Table 28: Phase B circuit gate. Conformal set construction fits the established budget at no measurable proving cost.

Head	logrows	Rows used	Proving time	Proof size
Temperature only	15	39	1.72 s	17,525 B
Temperature + conformal	15	95	1.66 s	17,828 B

It fits inside the established logrows = 15 budget, adding 56 rows of 32,768 and no measurable proving time. The reason is that both comparisons push through the monotone sigmoid into logit space and become comparisons against two constants — so the sigmoid never enters the circuit and the head stays affine. Conformal set construction is therefore **promoted to tier 1**, proved on the same terms as the confidence itself.

Two things this does not buy, stated because they are easy to overstate. The quantile q is not computed in-circuit: that needs a sort over calibration scores and it is a per-deployment constant, so it is fitted off-chain and committed exactly as T is — its provenance is exactly as unproven as T's. And the coverage guarantee is *marginal*: a conformal predictor can hit 90% overall while covering a protected subgroup at 60%, which is the same limitation aggregate ECE has (§8.4).

A.4 Counterfactual evidence

Table 29: Phase D. Counterfactuals over 40 rejections, checked for actionability and plausibility.

Metric	Value
Rejections examined	40
Counterfactual found	40 (100%)
Mean changes when found	1.075
Immutable-attribute violations	0
Implausible (unobserved) values	0
Mean overlap with SHAP top-5	0.762
Hash reproduces on rerun	True

The plausibility check earned its place on the first run, which flagged 10 of 40 cases: quantile-derived candidate values are interpolated, so the search was proposing loan amounts occurring nowhere in the data. "Reduce the loan to 3,271 DM" is not an actionable instruction when no such loan exists, and candidates are now snapped to observed values. The 0.762 overlap with the SHAP top-5 is a consistency check on the bundle, since the two explanations are committed together and routine disagreement would make the evidence internally inconsistent. The search is greedy, so what is reported is sparsity, not minimality.

A.5 Collusion detection, on synthetic rings only

A two-hop GraphSAGE classifier over the claimant projection of the dispute graph, against a degree-concentration baseline. Intensity is the fraction of a ring member's disputes aimed at the ring's operator.

Table 30: Phase E. Detection against injected rings only; intensity is the fraction of a ring member's disputes aimed at its own operator.

Ring intensity	GNN AUC	Degree-baseline AUC
1.00	1.000 ± 0.001	0.984 ± 0.009
0.85	0.980 ± 0.013	0.958 ± 0.014
0.70	0.884 ± 0.097	0.865 ± 0.102
0.60	0.856 ± 0.104	0.841 ± 0.104
0.50	0.807 ± 0.040	0.774 ± 0.066

At intensity 1.00 a ring is trivially separable and the AUC of 1.000 measures nothing but that separability, so the sweep continues to 0.50, where a ring member sends half its disputes elsewhere and the signal must come from graph structure.

This is the weakest evidence in the document, and it now carries the strongest consequences. Labels exist only because the rings were injected. There is no labelled real collusion — the protocol has never run — so no real-world detection performance is claimed or claimable, and the false-positive rate on genuine dispute traffic is unknown.

The detector nonetheless has an on-chain enforcement path (`CollusionOracle`). An enforced flag blocks a claimant from filing disputes and withholds their payout. This is a deliberate decision to give an unvalidated model authority over money, and §8.5 states what it costs.

Appendix B — Reproducing the results

Every number in this document is produced by a script in the repository; none is transcribed by hand. `10_train_calibrate.py` carries an ECE gate that fails loudly if calibration does not reduce error, so a silently broken run cannot produce a quiet result.

This sequence was executed against a fresh `git clone` on 4 September 2026, not merely asserted. Doing so found a real defect in an earlier version of this appendix: `contracts/lib/` is not committed, so `forge test` fails on a clean checkout with `Source "forge-std/Test.sol" not found`. The `forge install` line below was missing and is now present. With it, the clean checkout runs 144 Solidity and 98 Python tests green, and `market_model.json` and `subgroup_adversary.json` regenerate **byte-identical** to the artifacts behind §4 and §8.4.

Two limits on that verification, stated so it is not read as more than it is: it was run on the same machine and the same OS as development, so it establishes that the repository is self-contained, not that the pipeline is portable across platforms. And the zkML stages (`00_setup_tools.sh`, `30_export_onnx.py`, `31_zk_prove.py`) were not re-run in the clean checkout — they depend on an EZKL binary fetched at setup, so §7.3's proving figures are reproduced from the committed artifacts rather than from a fresh proving run.

```
pip install -r requirements.txt
bash scripts/00_setup_tools.sh          # EZKL CLI v23.0.5

# forge-std is NOT committed; without this, every Foundry command fails.
cd contracts && forge install foundry-rs/forge-std@v1.16.2 --no-git && cd ..

python scripts/10_train_calibrate.py    # §7.1  (ECE gate; fails loudly if not reduced)
python scripts/20_explain.py            # §7.2
python scripts/30_export_onnx.py        # ONNX export, fidelity-checked
python scripts/31_zk_prove.py           # §7.3  circuits, proofs, soundness
cd contracts && forge test              # §7.8  144 tests incl. ThreatModel.t.sol
python -m pytest tests/ -q              # §7.8  98 tests

anvil &                                  # §7.4  end-to-end
forge script script/Deploy.s.sol:Deploy --rpc-url http://127.0.0.1:8545 --broadcast
python scripts/40_demo_e2e.py
python scripts/90_render_results.py     # regenerate RESULTS.md
python scripts/91_figure_d_reliability.py   # Figure 4
python scripts/13_phaseA_ablation.py     # §A.1  ensemble ablation
python scripts/14_phaseB_conformal.py    # §A.3  conformal coverage
python scripts/15_phaseB_circuit_gate.py # §A.3  circuit feasibility gate
python scripts/16_phaseC_deep_ensemble.py # §A.2  epistemic uncertainty
python scripts/17_phaseD_counterfactual.py # §A.4  counterfactual evidence
python scripts/18_phaseE_collusion.py    # §A.5 collusion detection
python scripts/19_phaseH_ablation_report.py  # regenerate ABLATION.md

python scripts/92_render_svg.py             # Figures 1 and 3
python scripts/93_figure_e_sweep.py          # Figure 5

# --- v1: the agentic-payments work -------------------------------------
python scripts/21_subgroup_adversary.py  # §8.4  subgroup null + ECE noise floor
python scripts/22_market_model.py        # §4    welfare model, all of §4.3-§4.5
python scripts/23_second_dataset.py      # §8.6  replication on Taiwan default
python scripts/24_detector_fpr.py        # §8.5  detector FPR at the enforced threshold

cd contracts                             # §5.1  ERC-8004 mirror
forge test --match-contract ERC8004ReputationAdapter

cd ../x402-middleware && npm install     # §2.5  reference x402 integration
npm test                                 # starts anvil, deploys, gates
```

Every number in §4 comes from `22_market_model.py` and is written to `artifacts/calibration/market_model.json`; every number in §8.4 comes from `21_subgroup_adversary.py` and `artifacts/calibration/subgroup_adversary.json`. The unbonding arithmetic in §8.3 is derived in `docs/UNBONDING_PERIOD_JUSTIFICATION.md` from the statutory sources cited there, and its inputs — realised Brier 0.1758 and $r = 5\%$ — are §7.1 and §7.7 respectively.

§8.6's replication comes from `23_second_dataset.py` and `artifacts/calibration/second_dataset.json`; §8.5's detector figures from `24_detector_fpr.py` and `artifacts/ablation/detector_fpr.json`. The second dataset is fetched once from OpenML and cached as parquet, so a remote change cannot silently alter a published number.

The results that could erode have guards. `tests/test_subgroup_calibration.py` asserts the permutation control, the minimum group size and the monotonicity of the noise floor in n, so §8.4's null cannot be eroded by quietly shrinking a threshold. `tests/test_second_dataset.py` pins the replication against per-dataset retuning, and pins the detector's *unfavourable* numbers against a later change that evaluates at a friendlier threshold — a test suite that only defends good results is a ratchet in one direction.